

A POSITIVE FORMULA FOR VOLUMES OF FLAT UNITARY CONNECTIONS ON COMPACT ORIENTED SURFACES

QUENTIN FRANÇOIS, DAVID GARCÍA-ZELADA, THIERRY LÉVY, PIERRE TARRAGO

ABSTRACT. We provide a manifestly positive expression for the volume of flat $U(n)$ -valued connections on punctured compact oriented surfaces. This volume is obtained by summing volumes of explicit polytopes describing coloured honeycombs on a polygon, in the spirit of the work of Knutson and Tao describing the spectrum of the sum of two hermitian matrices. As a corollary, we also provide a positive formula for marginals of the $U(n)$ -valued Yang-Mills measure on a compact oriented surface in terms of the probability distribution of an explicit path process.

1. INTRODUCTION

Moduli spaces of flat unitary connections have been introduced by Narasimhan and Seshadri [NS65; Don83] as a way to classify vector bundles on a compact Riemann surface. As highlighted in the work of Atiyah and Bott [AB83], these moduli spaces are deeply related to the $U(n)$ -valued Yang-Mills measure on the corresponding surface. The latter is a random $U(n)$ -valued connection weighted by its curvature and depending on a temperature parameter. As the temperature vanishes, the only surviving connections are the ones with zero curvature and the zero-temperature limit of the Yang-Mills functional yields a natural symplectic volume form on the moduli space of flat connections.

Since the paper of Atiyah and Bott, there have been several works pursuing the goal of a better understanding of either the Yang-Mills measure or its zero temperature limit, the volume form on flat connections. Let us mention in particular two pioneering directions that appeared around 1990. The first one, initiated by Verlinde [Ver88], expresses the volume form as the limit of the dimension of vector spaces of holomorphic sections on the moduli space. This expression is clearly positive as a limit of positive integers. There is an explicit formula to compute these integers, originally due to Verlinde and later proved by Bismut and Labourie [BL99], involving sums of complex numbers in the case of the three-holed sphere. However, until recently [Buc+16], no manifestly positive formula existed for the dimension of such spaces.

The second work, due to Witten [Wit92], reduces the case of an arbitrary compact oriented surface to a three-holed sphere by exploiting a pants decomposition of the surface. Such a surgery procedure has since been rigorously proved several times, see [JK98] and [MW99a]. Using this method, Witten successfully obtained a general formula for the volume of flat connections as a series of characters of the unitary group. This first result, which is a series of complex numbers, has been improved to an alternating sum of positive numbers by Jeffrey [Jef94] and then Meinrenken and Woodward [MW99a].

In the present paper, we provide a positive formula for the volume of such moduli spaces as a sum of volumes of explicit finite dimensional polytopes. Denote by $M_{g,p}(\alpha_1, \dots, \alpha_p)$ the moduli space of flat unitary connections on an oriented punctured compact surface of genus g with p removed points, such that the holonomy around the i -th point belongs to an orbit $\alpha_i \subset U(n)$ which is supposed to be non-degenerate. Then, our result may be informally stated as follows, see Theorem 2.6 for a precise statement:

$$\text{Vol}[M_{g,p}(\alpha_1, \dots, \alpha_p)] = \left(\frac{2^{(n+1)[2]} (2\pi)^{(n-1)(n-2)}}{n!n} \right)^{2g+p-2} \frac{n}{\prod_{j=1}^p \Delta(\alpha_j)} \sum_{G \in \mathcal{G}^{(g,p)}} \text{Vol}[P_{\alpha_1, \dots, \alpha_p}^G],$$

where Δ is an explicit function involving a Vandermonde determinant and $\{P^G, G \in \mathcal{G}^{(g,p)}\}$ is a collection of explicit polytopes indexed by a family of graphs $\mathcal{G}^{(g,p)}$ and whose boundaries depend on the values of $\alpha_1, \dots, \alpha_p$. These polytopes consist of coordinates of a collection of segments on a polygonal surface, see Figure 3 for a typical example of such configurations. The surface itself simply reflects the pants decomposition of the original punctured oriented surface, while the collection of segments is a colored generalization of the honeycomb model introduced by Knutson and Tao [KTW04] to describe the spectrum of the sum of two hermitian matrices with prescribed spectra.

The fact that our model is a generalization of [KTW04] should not be a surprise, since the solution of the eigenvalue problem for the sum of hermitian matrices yields a particular subexample of a flat connection on a three-holed sphere, see [Kly98; Bel08]. Actually, this subexample has been used to get the previously mentioned signed formulas for the volume of the moduli spaces on the three-holed sphere, see [MW99a].

The proof of the present result is mainly based on the positive formula given in [FT24] in the case of the three-holed sphere. Roughly speaking, this formula had been obtained by following the initial method of Verlinde together with the recent results on the computation of the structure constants of the quantum cohomology of the Grassmannians [Buc+16]. In [FT24], the formula was given in terms of volumes of a so-called coloured hive model, which consists of a union of some explicit but degenerate polytopes.

Note that volumes are always defined up to a constant and when polytopes of a Euclidean space are degenerate, there are several ways to choose this constant from the volume measure of the ambient space: one can either consider the volume induced by the restriction of the Riemannian metric or consider the pull-back of the canonical Lebesgue measure by a bijective projection on a subset of a canonical basis. Both definitions agree in the case of a convex body (that is, a convex polytope of the same dimension as the ambient space), but they start to differ when degeneracies appear. The choice of a volume form was relatively inconsequential in the case of [KTW04], since there was only one polytope considered and the change of volume form only amounted to a change by an overall constant. However, we are considering here a sum of volumes of polytopes that are degenerate and each lie in different subspaces of the ambient space, so that we must ensure that the chosen volumes are coherent from one polytope to the other. In [FT24], the correct choice was to use the pull-back corresponding to particular bijective projections.

It appears that the formulation of [FT24] in terms of hives does not fit easily in the surgery formulas of Witten. In order to generalize the case of the three-holed sphere to arbitrary punctured compact oriented surfaces, our first task has been to reformulate the latter hive model in terms of a colored version of honeycombs. As a byproduct of this reformulation, we also obtain a new volume formula in the case of the three-holed sphere which is coordinate-free : we provide an explicit relation between the volume form induced by the Riemannian metric of the ambient space and any volume induced by a bijective projection onto some coordinates. This relation is given by a simple combinatorial number, the number of spanning trees of a graph, in a similar spirit to [KL22]. Thanks to the new honeycomb model, the surgery formulas of [Wit92] to compute the volume of flat connections easily translate into a patching of honeycombs, and we thus get the desired positive formula for surfaces of arbitrary genus.

Note that this formula suggests a simple geometric description of a flat $U(n)$ -connection on a compact oriented surface, up to an overall conjugation by a unitary matrix, by the corresponding honeycomb arrangement. It would be very interesting to provide an explicit measure preserving bijection between the two models. However up to now, such a bijection has been out of reach already for the simplest particular case of the sum of hermitian matrices.

As suggested by [AB83], volumes of flat connections are the building blocks for the computations of the partition function of the two-dimensional Yang-Mills measure on compact oriented surfaces, see [Wit92, Section 2]. Using the new positive formulas for flat connections, one can thus easily deduce new formulas for marginals of the Yang-Mills measure which are again manifestly positive. Remark that our formulas are only valid for the holonomy on a set of non-crossing curves, and this thus gives only a partial description of the full Yang-Mills measure. Once again, our result suggests the existence of a measure preserving bijection between a projection of the Yang-Mills measure and random path processes consisting of Dyson Brownian motions on the circle and certain frozen path configurations in polygons, see Remark 2.9.

2. NOTATIONS AND STATEMENT OF THE MAIN RESULT

2.1. Conjugacy classes of the unitary group. Throughout this paper, we fix an integer $n \geq 3$ and denote $\mathcal{H} = (\mathbb{R}/\mathbb{Z})^n / S_n$, where S_n is the symmetric group of degree n acting on $(\mathbb{R}/\mathbb{Z})^n$ by permutations of the coordinates. As a set, we identify $\mathcal{H}$ with $\{\theta = (\theta_1, \dots, \theta_n) \in [0, 1]^n : \theta_1 \geq \dots \geq \theta_n\}$ in the usual way. For our purposes, $\mathcal{H}$ represents the set of conjugacy classes of $U(n)$, where the

conjugacy class of $\theta \in \mathcal{H}$ is

$$\mathcal{O}(\theta) = \left\{ U e^{2\pi i \theta} U^{-1} \mid U \in \mathrm{U}(n) \right\} \quad \text{with} \quad e^{2\pi i \theta} = \begin{pmatrix} e^{2\pi i \theta_1} & 0 & \dots & 0 \\ 0 & e^{2\pi i \theta_2} & & \vdots \\ \vdots & & \ddots & \\ 0 & \dots & & e^{2\pi i \theta_n} \end{pmatrix}.$$

Let us use the notation $\mathcal{H}_{reg} = \{\theta \in \mathcal{H} \mid \theta_1 > \theta_2 > \dots > \theta_n\}$ which represents the set of regular conjugacy classes of $\mathrm{U}(n)$, namely the ones of maximal dimension in $\mathrm{U}(n)$. Finally, let $\mathcal{H}_{reg}^0$ denote the subset of $\mathcal{H}_{reg}$ corresponding to the regular conjugacy classes of $\mathrm{SU}(n)$. Namely,

$$\mathcal{H}_{reg}^0 = \left\{ \theta \in \mathcal{H} \mid \theta_1 > \theta_2 > \dots > \theta_n \text{ and } \sum_{i=1}^n \theta_i \in \mathbb{N} \right\}.$$

Consider the moduli space $\mathcal{M}_{g,n}(\alpha_1, \dots, \alpha_p)$ of flat $\mathrm{U}(n)$ -valued connections on a compact oriented surface with genus g having p boundary components $\mathcal{L}_1, \dots, \mathcal{L}_p$, for which the holonomy around each $\mathcal{L}_i$, $1 \leq i \leq p$, belongs to $\mathcal{O}_{\alpha_i}$. Then, there is a symplectic structure on the quotient space $M_{g,n}(\alpha_1, \dots, \alpha_p) = \mathcal{M}_{g,n}(\alpha_1, \dots, \alpha_p) / \mathcal{G}_n$, where $\mathcal{G}_n$ is the $\mathrm{U}(n)$ -valued gauge group on the surface, see [MW99b]. The space $M_{g,n}(\alpha_1, \dots, \alpha_p)$ inherits thus a canonical volume form from the symplectic form.

The main goal of the present paper is to provide an explicit and positive formula for the corresponding volume of $M_{g,n}(\alpha_1, \dots, \alpha_p)$. This formula is given in terms of certain configurations of lines, called honeycomb, which are a generalization of the honeycomb model introduced in [KTW04] and are described in the next two paragraphs.

2.2. Triangular honeycomb. Let us first define a triangular honeycomb, which will contribute to the volume of the moduli space in the case $g = 0, p = 3$, see Theorem 4.16.

Consider $S \subset \mathbb{R}^2$, a convex subdomain of the plane. Recall that a segment of S is any set of the form $\{xv + (1-x)v', 0 \leq x \leq 1\}$ for some $v, v' \in S$. For any such segment $e = \{xv + (1-x)v', 0 \leq x \leq 1\}$, we then set $\partial e = \{v, v'\}$ and denote by $\overset{\circ}{e} = e \setminus \{v, v'\}$ its interior. A segment is then called non-trivial if $\overset{\circ}{e} \neq \emptyset$. Denote by $\mathbb{G}$ the set of non-trivial segments of S .

If e, e' are two non-trivial segments, we define the angle from e to e' as

$$\widehat{(e, e')} = \arccos(\langle u, u' \rangle),$$

where u, u' are unit tangent vectors of respectively e and e' such that $\det(u, u') > 0$.

Definition 2.1 (Non-degenerate honeycomb). A *honeycomb* $h = (\mathcal{E}, c)$ is a finite subset $\mathcal{E} \subset \mathbb{G}$ together with a color map $c : \mathcal{E} \rightarrow \{0, 1, 3\}$ such that :

- (1) $\overset{\circ}{e} \cap \overset{\circ}{e'} \neq \emptyset$ is only possible if, up to a transposition of e and e' , $c(e) = 0, c(e') = 1$ and $\widehat{(e, e')} = \pi/3$,
- (2) if $\partial e \cap \partial e' = \{v\}$, then $(c(e), c(e')) \in \{(0, 0), (1, 1), (0, 1), (1, 3), (3, 0)\}$ and $\widehat{(e, e')} = 2\pi/3$.

From the definition that $\mathcal{E}$ is unambiguously defined from the representation $\bigcup_{e \in \mathcal{E}} e$ of h as a subset of S . In the sequel, we set

$$t(h) = \bigcup_{e \in \mathcal{E}} e.$$

Figure 1 lists the possible angles between segments in non-degenerate honeycombs of Definition 2.1. Remark that our choice of colors is coherent with the colors coming from [Buc+16] and [FT24] : the colors 0 and 1 play a symmetric role that differs from the one played by the color 3.

Definition 2.2 (Structure graph). The *structure graph* of a honeycomb h is the finite graph $G(h) = (V, E)$ with colored edges such that $V = \bigcup_{e \in \mathcal{E}} \partial e$, $E = \{\partial e, e \in \mathcal{E}\}$, and the color map $c : E \rightarrow \{0, 1, 3\}$ is defined by $c(v, v') = c(e)$ if $\partial e = \{v, v'\}$.

By the angle condition, $G = G(h)$ has only vertices of degree less than 3 and the sequence of colors around any trivalent (resp. bivalent) vertex of $G(h)$ belongs to $\{(0, 0, 0), (1, 1, 1), (0, 3, 1)\}$ (resp. is equal to $(0, 1)$) in the clockwise order.

The *color number* $c(G)$ of a colored graph G is defined as the number of edges colored 1 and adjacent

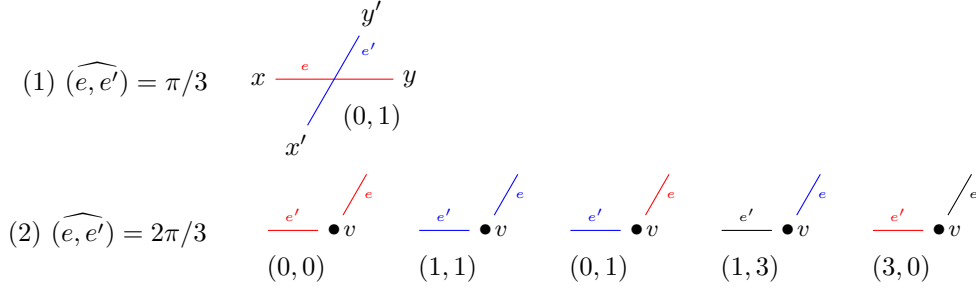


FIGURE 1. The possible angles between segments. The colors of the edges are given in the couple below the configuration.

to a univalent vertex. By abuse of definition, we speak of edges and vertices of a honeycomb to denote edges and vertices of its canonical structure graph.

Let us denote by $T := \{x + ye^{i\pi/3} \mid 0 \leq x, y \leq 1, x + y \leq 1\} \subset \mathbb{C}$ the equilateral triangle with vertices $0, 1$ and $e^{i\pi/3}$. To each point $v \in T$ we associate the triple (v_0, v_1, v_2) such that $v = v_1 + v_2e^{i\pi/3}$ and $v_0 = 1 - v_1 - v_2$. Then, the boundary ∂T can be decomposed as

$$\partial T = \bigsqcup_{i \in \{0,1,2\}} \partial_i T, \text{ where } \partial_i T = \{v \in T \mid v_i = 0\}.$$

Definition 2.3 (Triangular honeycomb). For $n \geq 1$, a *triangular honeycomb* h of size n is a non-degenerate honeycomb $h = (\mathcal{E}, c)$ on the surface T endowed with the Euclidian metric such that

- (1) $G(h)$ has only univalent and trivalent vertices, and $t(h) \cap \partial T = \partial V$, where ∂V denotes the set of univalent vertices in $G(h)$,
- (2) if $e \in \mathcal{E}$, then $e \subset \{x + \mathbb{R}e^{2\pi i(\ell(e)+1)/3}\}$ for some $\ell(e) \in \{0, 1, 2\}$ and $x \in T$. The integer $\ell(e)$ is then called the *type* of e and $L(e) = x_{\ell(e)}$ is called the *height* of e which is independent of the choice of x ,
- (3) for $0 \leq i \leq 2$, $\#t(h) \cap \partial_i T = n$ and if e is adjacent to a boundary vertex belonging to $\partial_i T$, then either $c(e) = 0$ and $\ell(e) = i + 1$ or $c(e) = 1$ and $\ell(e) = i + 2$. Moreover, the color is increasing along each edge : namely, if e^1 (resp. e^2) meets $\partial_i T$ at x^1 (resp. x^2) with $x_{i+1}^2 > x_{i+1}^1$, then $c(e^2) \geq c(e^1)$.

A triangular honeycomb has always $n^2 + 3n$ vertices and $\frac{3n(n+1)}{2}$ edges, see Remark 4.15. This definition implies that $G(h) = (V, E)$ has only trivalent vertices in $\overset{\circ}{T}$. Moreover, the condition on the boundaries yields a natural choice of root v^0 of G , corresponding to the univalent vertex lying on $\partial_0 T$ and whose coordinate v_1^1 is maximal. Then, the cyclic counter-clockwise order on the boundary vertices given by the orientation of T yields an order on the $3n$ boundary vertices $\{v^1, \dots, v^{3n}\}$ of ∂V , with the vertices $\{v^{ni+j}, 1 \leq j \leq n\}$ located on $\partial_i T$.

In the particular case where all edges have the same color, our definition is the original definition of a generic honeycomb from [KT99]. Remark that, besides the coloring, our definition of triangular honeycombs differs slightly from the original one since we impose vertices to be trivalent inside $\text{int}(T)$. The set of honeycombs in their original definition can then be seen as the closure of the ones from the present manuscript (in the case where all colors are the same).

The *boundary*, or the *boundary values*, of a triangular honeycomb h of size $n \geq 3$ is the $3n$ tuple

$$(2.1) \quad \partial h := ((\alpha_n^0 \leq \dots \leq \alpha_1^0), (\alpha_n^1 \leq \dots \leq \alpha_1^1), (\alpha_n^2 \leq \dots \leq \alpha_1^2)),$$

where $(\alpha_n^i \leq \dots \leq \alpha_1^i)$ is the ordered tuple of the $i + 1$ -coordinates of boundary points of $t(h)$ on $\partial_i T$. Hence, $\alpha_j^i = v_{i+1}^{ni+j}$ for $0 \leq i \leq 2$ and $1 \leq j \leq n$.

For $\alpha, \beta, \gamma \in \mathcal{H}_{reg}$ let us denote by $\text{HONEY}_{n,d}(\alpha, \beta, \gamma)$ the set of triangular honeycombs h having boundary values $\partial h = (\beta, \alpha, \gamma)$, see Figure 2, and such that there are d edges colored 1 meeting one (and thus each) boundary component of T . For any colored graph G with an order on the boundary

vertices, let us denote by

$$\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)$$

the set of triangular honeycombs of $\text{HONEY}_{d,n}(\alpha, \beta, \gamma)$ with canonical graph structure isomorphic to G as colored graph with ordered boundary. Then, if $\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)$ is non-empty, necessarily the graph G has $3d$ univalent vertices adjacent to edges colored 1. We then define the color of G as $c(G) = d$. Let us denote by $\mathcal{G}_d$ the set of isomorphism classes of colored graphs with ordered boundary appearing in $\{G(h) \mid h \in \text{HONEY}_{n,d}\}$.

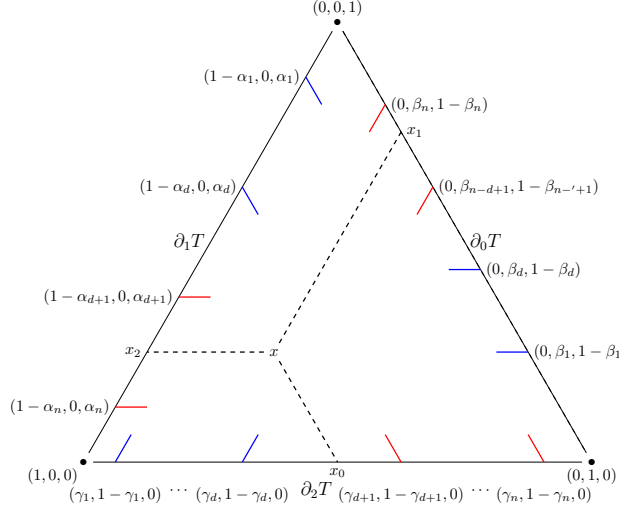


FIGURE 2. Boundaries of a honeycomb in $\text{HONEY}_{n,d}(\alpha, \beta, \gamma)$.

2.3. (g, p) -honeycombs. We next turn to a generalization of triangular honeycombs which will provide a formula for arbitrary surfaces. Let $g, p \geq 0$ be integers such that $N = p + 2g - 2 \geq 1$ and let S be a connected compact orientable surface of genus g with p boundary components $\mathcal{L}_1, \dots, \mathcal{L}_p$. Let $\mathcal{M} = (M_1, \dots, M_N)$ be a pant decomposition of this surface. We build from $\mathcal{M}$ a surface $\mathcal{T}$ as follows:

- take N oriented equilateral triangles $(T^1, \dots, T^N)$, each T^i having three oriented boundaries $\partial_j T^i$ of size 1.
- For each pair of boundary components $\partial_j M^i, \partial_{j'} M^{i'}$ which are identified in the pair of pants decomposition, write $(i, j) \sim (i', j')$ and identify the boundaries $\partial_j T^i$ and $\partial_{j'} T^{i'}$ in an isometric and orientation reversing way.

Then, the resulting surface $\mathcal{T}$ is an oriented surface with p boundaries edges denoted by $L_1, \dots, L_p$. If L_i a boundary component, there exists a triangle T_{j_i} and $\ell_i \in \{0, 1, 2\}$ such that $L_i = \partial_{\ell_i} T^{j_i}$.

For each $1 \leq i \leq p + 2g - 2$, let $f_i : T \rightarrow \mathcal{T}$ be an orientation preserving isometry such that $f_i(T) = T^i$ (where T^i is seen as a subset of $\mathcal{T}$ by the above construction). For a triangular honeycomb $h = (\mathcal{E}, c)$, set $t_i(h) = \bigcup_{e \in \mathcal{E}} f_i(e)$.

Definition 2.4 ((g, p) -honeycomb). A (g, p) -honeycomb is a tuple $(h^i)_{1 \leq i \leq N}$ of triangular honeycombs such that $t_i(h^i)|_{\partial_j T^i} = t_{i'}(h^{i'})|_{\partial_{j'} T^{i'}}$ for any $(i, j) \sim (i', j')$.

See Figure 3 for the representation of a (g, p) -honeycomb for $g = 2$ and $p = 3$.

Remark 2.5. The natural euclidean metric on each equilateral triangle yields a metric on $\mathcal{T}$ which is flat except at the vertices of the triangulation belonging to the interior of $\mathcal{T}$. Hence, writing $h^i = (\mathcal{E}^i, c^i)$ for $1 \leq i \leq 2p + g + 2$, $\bigcup_{i=1}^N \mathcal{E}^i$ defines again a non-degenerate honeycomb on $\mathcal{T}$, where we extended Definition 2.1 from a subdomain of $\mathbb{R}^2$ to any locally flat Riemannian oriented surface.

Let us denote by $\text{HONEY}^{(g,p)}$ the set of (g, p) -honeycombs. For $h = (h^i)_{1 \leq i \leq N} \in \text{HONEY}^{(g,p)}$ and $1 \leq i \leq N$, let $G(h^i) = (V^i, E^i)$ be the structure graph of the honeycomb h^i with corresponding color map $c_i : E^i \rightarrow \{0, 1, 3\}$. We then define the structure graph of h as the graph $G(h) = (V, E)$ where

$$V = \bigcup_{1 \leq i \leq N} f_i(V^i) \quad \text{and} \quad E = \bigcup_{1 \leq i \leq N} \{f_i(e), e \in E^i\}.$$

We define a coloring of the edges of this graph by setting $c(f_i(e)) = c_i(e_i)$ if $e \in E^i$.

If h is a (g, p) honeycomb and $1 \leq i \leq p$, we set $\partial_i h = t(h) \cap L_i$. Each element v of $t(h) \cap L_i$ belongs to a unique triangle T^{ji} and inherits a triple of coordinate (v_0, v_1, v_2) from this triangle. Moreover, as in the triangular case, the orientation of S yields an orientation on each boundary component. For $h \in \text{HONEY}^{(g,p)}$, order the boundary points so that

$$t(h) \cap \bigcup_{1 \leq i \leq p} L_i = \{v^{(i-1)n+j}, 1 \leq i \leq p, j \leq n\},$$

where $\partial_i h = \{v^{(i-1)n+j}, 1 \leq j \leq n\}$ with $v_{\ell_i+1}^{(i-1)n+j} > v_{\ell_i+1}^{(i-1)n+j'}$ if $1 \leq j < j' \leq n$. Hence, the structure graph $G(h)$ inherits the previous order on its boundary points.

We denote by $\text{HONEY}^{(g,p)}(\alpha^1, \dots, \alpha^p)$ the set of (g, p) -honeycombs with boundary components $\partial_i h = \alpha^i$. Like in the triangular case, we denote by HONEY^G the subset of $\text{HONEY}^{(g,p)}$ of honeycombs h with structure graph $G(h)$ isomorphic to G as colored graph with ordered boundary vertices. We also denote by $\mathcal{G}^{(p,g)}$ the set of isomorphism classes of colored graphs with ordered boundary appearing as structure graph of elements of $\text{HONEY}^{(g,p)}$. For $G \in \mathcal{G}^{(g,p)}$, let us set

$$c(G) = \sum_{i=1}^N c(G_i),$$

where $c(G_i)$ is the color number of the structure graph G_i as defined at the end of Section 2.2. Let us denote by $\mathcal{G}_d^{(g,p)} \subset \mathcal{G}^{(g,p)}$ the subset of graphs G such that $c(G) = d$.

For $G = (V, E) \in \mathcal{G}^{(p,g)}$, there is a natural parametrization of HONEY^G constructed as follows. For any edge $e \in E$, there exists $1 \leq i \leq N$ such that $e = f_i(\tilde{e})$ with $\tilde{e} \in E^i$. For $h \in \text{HONEY}^G$, let us set

$$P[h](e) = L(\tilde{e}),$$

where L is the height map defined on T from Definition 2.3. Then, introduce the map

$$(2.2) \quad P : \text{HONEY}^G \rightarrow \mathbb{R}^E,$$

sending $h \in \text{HONEY}^G$ to $(P[h](e))_{e \in E}$. This map is clearly non-surjective since $P(\text{HONEY}^G)$ is a bounded subset of $(\mathbb{R}^{3n(n+1)/2})^N$. Moreover, there are several relations among the values of $((P[h](e))_{e \in G_h^i})_{1 \leq i \leq N}$ which imply that P is an over-parametrization of HONEY^G . However, it will follow from Proposition 4.3 that P is injective. In the sequel, we identify HONEY^G with its image through P .

As a consequence of the decomposition described in Section 5 below, all graphs of $\mathcal{G}^{(g,p)}$ have the same number $n_e = \frac{3Nn(n+1)}{2}$ of edges and $n_v = Nn^2 + \frac{(3N-(p+2g))n}{2}$ of vertices. Hence, up to identifying edges and vertices of these graph, one can assume that there is a unique set $E^{(g,p)}$ of edges (resp. set $V^{(g,p)}$ of vertices) common to all graphs of $\mathcal{G}^{(g,p)}$ and that the graph structure of $G \in \mathcal{G}^{(g,p)}$ is encoded in the map $\partial : E^{(g,p)} \rightarrow \mathcal{P}(V^{(g,p)})$ which associates to an edge its endpoints.

2.4. Volume formula for the moduli space of flat connections. For $g, p \geq 0$, Set $d\text{Vol} \in \Omega_{g,p}^{n_{g,p}}(\mathbb{R}^{E^{(g,p)}})$ for the volume form on $n_{g,p}$ -dimensional subspaces induced by the canonical Riemannian metric on $\mathbb{R}^{E^{(g,p)}}$. Hence,

$$(2.3) \quad d\text{Vol} = \sum_{1 \leq i_1 < \dots < i_{n_{g,p}} \leq n_e} dx_{i_1} \wedge \dots \wedge dx_{i_{n_{g,p}}}.$$

In the following statement, we write $\text{Vol}(K)$ for the integration of $d\text{Vol}$ on a $n_{g,p}$ -dimensional submanifold and we let $\Delta(x) = 2^{n(n-1)/2} \prod_{i < j} \sin(\pi(x_i - x_j))$ denote the absolute value of the Vandermonde determinant of $(e^{ix_j})_{1 \leq j \leq n}$.

Let $Z_{g,p}(\alpha_1, \dots, \alpha_p)$ be the volume function for the moduli space of flat unitary connections on a compact oriented surface with genus g and with p boundary components around which the holonomies belong to $\alpha_1, \dots, \alpha_p$. Since the product of the determinant of the holonomies around the removed point is the identity, $Z_{g,p}(\alpha_1, \dots, \alpha_p)$ is non-zero only if $\sum_{i=1}^p |\alpha_i| \in \mathbb{N}$, where $|x| = \sum_{j=1}^n |x_j|$ for $x \in \mathbb{R}^n$.

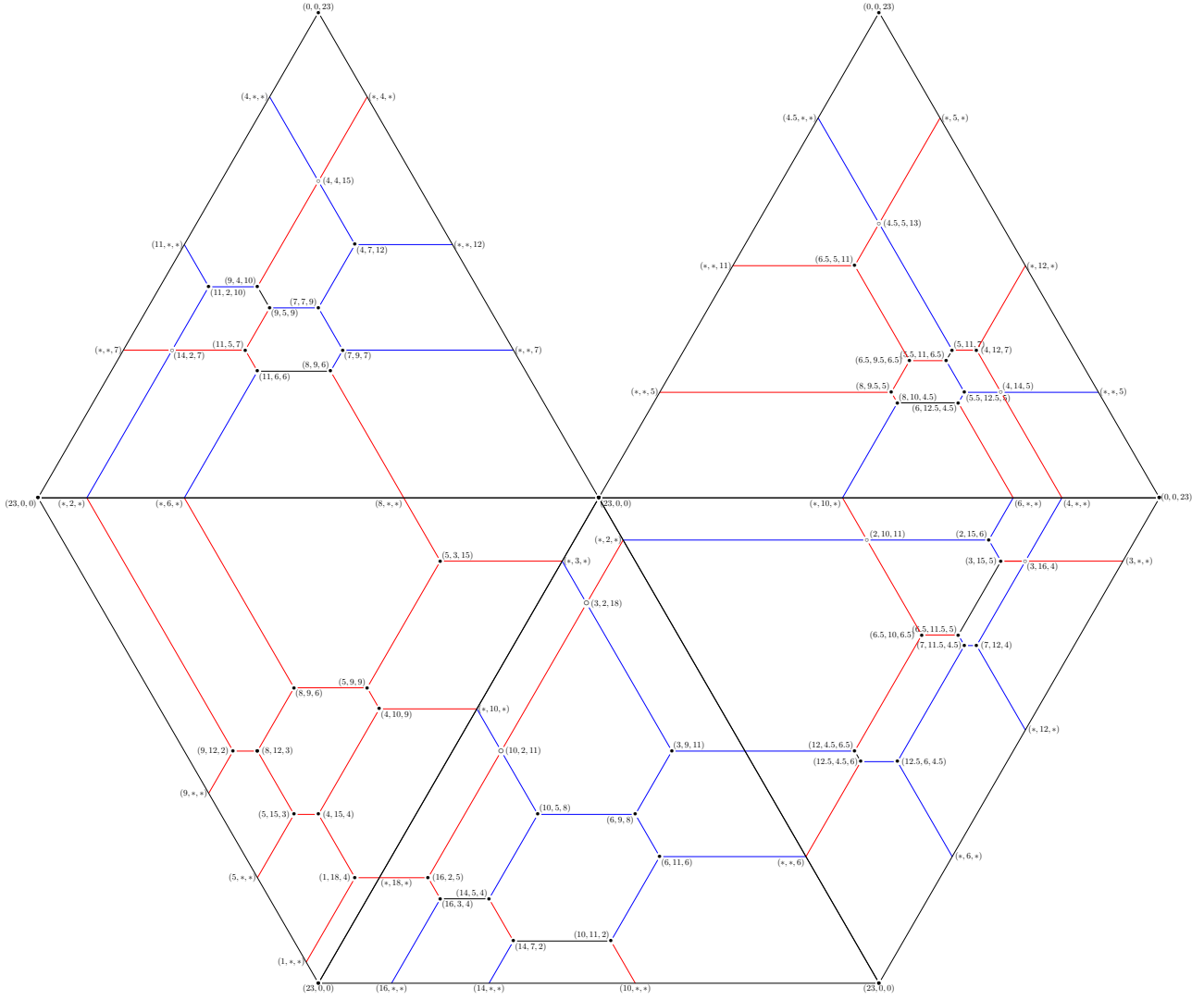


FIGURE 3. A $(2,3)$ honeycomb for a surface with genus 2 and 3 boundaries. On each upper triangle, the endpoints of the segments of the honeycomb coincide on the left and on the right edge.

Theorem 2.6 (Volume formula for (g,p) partition function). *For $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{reg}$ such that $\sum_{i=1}^p |\alpha_i| \in \mathbb{N}$,*

$$(2.4) \quad Z_{g,p}(\alpha_1, \dots, \alpha_p) = \frac{c_{0,3}^N}{n^{2g+p-3} \prod_{j=1}^p \Delta(\alpha_j)} \sum_{G \in \mathcal{G}^{(g,p)}} \frac{\text{Vol}[\text{HONEY}^G(\alpha_1, \dots, \alpha_p)]}{\sqrt{\# \text{ Spanning trees of } G}},$$

$$\text{where } c_{0,3} = \frac{2^{(n+1)[2]} (2\pi)^{(n-1)(n-2)}}{n!}.$$

The sum on the right hand-side is finite since the set $\mathcal{G}^{(g,p)}$ is finite. As it will appear below, for given $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{reg}$ the volumes appearing in the sum will be non-zero only for a subset $G \in \mathcal{G}_d^{(g,p)}$ where $d = \sum_{i=1}^p \sum_{j=1}^n \alpha_j^i - (p-2)n$.

The formula of Theorem 2.6 also yields a formula for $\text{SU}(n)$ -valued connections, since the volume for the $\text{SU}(n)$ case is equal to the one of the $\text{U}(n)$ case for $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{reg}^0$, see (6.2).

There are several other conventions to define the volume of flat connections, depending on the chosen volume on the unitary group (see [Wit92, Section 4.1]) and depending on the kind of volume form considered. On the latter convention, remark that we are choosing here the symplectic volume, more adapted to Yang Mills generalizations, rather than the torsion volume introduced in [Wit92], see also [MW99a, Appendix A] for a symplectic point of view on the relation between both volumes. The volume forms are easily related by the explicit factor $\prod_{j=1}^p \Delta(\alpha_j)$, see [Wit92, Section 4.7].

Remark 2.7 (Volume and conditioned processes). For each $G \in \mathcal{G}^{(g,p)}$, choosing a spanning tree $S \subset E^{(g,p)}$ of G yields a natural parametrization of $\text{HONEY}^G(\alpha_1, \dots, \alpha_p)$ in terms of the coordinates $P(e), e \in S$. By construction, each of those coordinates belongs to $[0, 1]$. Considering the uniform probability measure on $[0, 1]^S$, one can prove that the formula $\frac{\text{Vol}[\text{HONEY}^G(\alpha_1, \dots, \alpha_p)]}{\sqrt{\# \text{Spanning trees of } G}}$ is then exactly the probability that the vector $(L(e))_{e \in S}$ belongs to some convex set K_G , see Lemma 5.2 for a definition of K_G . After the summation on all graphs G , $Z_{g,p}(\alpha_1, \dots, \alpha_p)$ can be seen as the probability that a random process remains in a union of convex set, up to an explicit constant. Visually, the non-conditioned process would allow segments to exit the domain and to meet at the trivalent vertices with angles being any multiple of $\pi/3$.

2.5. Yang-Mills marginal for disjoint curves. As a consequence of Theorem 2.6, there is an explicit formula for the marginal of $U(n)$ -valued Yang–Mills partition function of an oriented surface of genus g with prescribed non-degenerate holonomies (up to conjugation) on a finite set of disjoint loops. This formula is given in Corollary 2.8 below. As it is proven in [Lév03], the partition function only depends on the prescribed conjugacy classes and on the areas of each connected components delimited by the loops.

Let S be a connected compact oriented surface of genus $g \geq 0$ together with p disjoint Jordan curves $\Gamma_1, \dots, \Gamma_p$ on S . For each Γ_i , $1 \leq i \leq p$, let α_i be an element of $\mathcal{H}_{\text{reg}}$.

We associate to $(S, \Gamma_1, \dots, \Gamma_p)$ a labeled finite directed graph $\mathcal{T} = (V, E)$ such that vertices are labelled by $\mathbb{R}_{\geq 0} \times \mathbb{N}$ and directed edges are labelled by $\mathcal{H}_{\text{reg}}$ as follows :

- the set V of vertices of $\mathcal{T}$ is the set of connected components of $S \setminus \bigcup_{i=1}^p \Gamma_i$. Each vertex $v \in V$ is labelled (A_v, g_v) where A_v is the area of the corresponding connected component and g_v is its genus.
- For $v_1, v_2 \in V$, there is a directed edge e from v_1 to v_2 if v_1 and v_2 are boundary components of a loop, and we write $v = s(e)$. We label the corresponding oriented edge e by α_e , with the condition that $\alpha_{\bar{e}} = 1 - \alpha_e$ if $\bar{e}$ is the oriented edge (v_2, v_1) .

We refer to Figure 4 for an example of such a configuration of loops and its corresponding edge-labelled tree. In the following, let us denote by d_v the degree of a vertex $v \in V$.

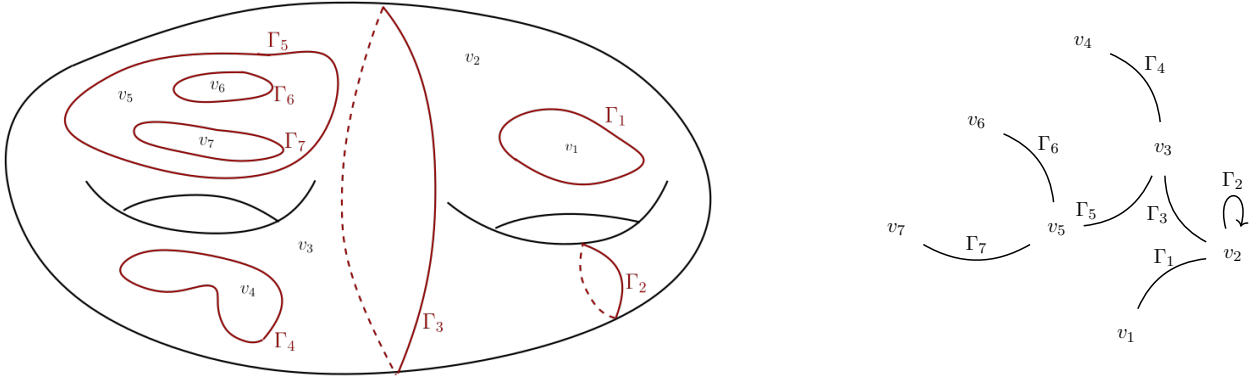


FIGURE 4. A disjoint loops configuration and its corresponding edge-labelled tree

For $T \geq 0$ and $x, y \in \mathcal{H}$, let us denote by $k_T(x, \cdot)$ the transition kernel from x to y of the unitary Dyson Brownian motion on $\mathcal{H}$, which is the projection on $\mathcal{H}$ of the $U(n)$ -valued Brownian motion starting at $\mathcal{O}(x)$, see (7.1).

Corollary 2.8 (Yang-Mills partition function). *The Yang-Mills partition function associated to the data $(S, \Gamma_1, \dots, \Gamma_p)$ is*

$$(2.5) \quad \text{YM}(\alpha_1, \dots, \alpha_p) = \int_{\mathcal{H}^{\sum_{i=1}^p d_v}} \prod_{v \in V} Z_{g_v, d_v}(\mu_1^v, \dots, \mu_{d_v}^v) \prod_{e \in E, v=s(e)} k_{A_v/d_v}(1 - \mu_i^v, \alpha_e) d\mu_i^v,$$

where $Z_{0,1}(\mu) = \delta_0$ and $Z_{g,p}$ is given in (2.4) otherwise.

Remark 2.9 (Yang-Mills marginals and conditioned processes). As a consequence of Theorem 2.6 and Remark 2.7, the latter formula expresses the partition function $\text{YM}(\alpha_1, \dots, \alpha_p)$ as the probability that

a random process remains in some domain and has the right boundary values. This random process consists in concatenations of Brownian motions on circle and straight paths in some frozen domains and the domain of restriction corresponds to the condition of being non-crossing for the Brownian motion and yielding (g, p) -honeycombs in the frozen domains. It would be interesting to see whether such interpretation allows simulations of a generical process contributing to the formula (2.5).

Organisation of the paper. In Section 3.1, we introduce a volume form on flows of graphs, together with formulas for two operations on graphs which we call sieving and contractions in Section 3.2. Section 4 is devoted to the proof of Theorem 2.6 in the case of the three-holed sphere which corresponds to $(g, p) = (0, 3)$. We first show that triangular honeycombs can be viewed as flows on graphs in Section 4.1. Section 4.2 recalls the *dual hive* model introduced in [FT24]. Sections 4.3 and 4.4 establish a volume preserving map from dual hives to triangular honeycombs. Using the result from [FT24] which expresses the partition function for the three-holed sphere in terms of volume of hives, we prove the volume formula of Theorem 2.6 in terms of triangular honeycombs. In Section 5, we establish two formulas corresponding to Proposition 5.3 and Proposition 5.4 which express the volume of honeycombs constructed from the operations of Section 3.2. The full proof of Theorem 2.6 is done in Section 6. We first give formulas relating the volume of moduli spaces of flat connections for surfaces glued together in Section 6.2 before proving the theorem in Section 6.3 by relating the previous formulas with the ones of Section 5. Finally, we prove Corollary 2.8 in Section 7.

3. GRAPHS, DIVERGENCE OF FLOWS AND VOLUME MEASURES

We will see a honeycomb as graph G endowed with a flow (an antisymmetric function on its edge set). The honeycomb information of G , i.e., the precise drawing on the equilateral triangle of size 1, will be obtained once we fix the distance of each edge to the side of the corresponding triangle parallel to it, which are embodied by the map L . It is thus important to give an explicit expression of the volume measure associated to this distances, which is the main objective of this section.

Let us consider a finite graph $G = (V, E)$ and let us denote $\vec{E} = \{(a, b) \in V \times V : \{a, b\} \in E\}$ its oriented edge set. A function $\omega : \vec{E} \rightarrow \mathbb{R}$ is called antisymmetric if

$$\omega(a, b) = -\omega(b, a).$$

Let us denote by $\Omega^0(G)$ the vector space of real functions on V endowed with the inner product given by $\langle f_1, f_2 \rangle = \sum_{x \in V} f_1(x) f_2(x)$. The space of flows is the vector space $\Omega^1(G)$ of antisymmetric real functions ω on $\vec{E}$ endowed with the inner product $\langle \omega_1, \omega_2 \rangle = \sum_{e \in E} \omega_1(e) \omega_2(e)$, where we are using that, due to the antisymmetry of ω_1 and ω_2 , the product $\omega_1(e) \omega_2(e)$ does not depend on the orientating of e as long as we choose the same for both arguments. For any map $\phi : V \rightarrow \mathbb{R}$, set

$$(3.1) \quad \mathcal{F}(\phi) = \left\{ \lambda \in \Omega^1(G) \mid \forall v \in V : \sum_{x \sim v} \lambda(v, x) = \phi(v) \right\}.$$

We will recall some properties of the *divergence* operator used in the previous equation and explore the space of solutions to provide a convenient measure on $\mathcal{F}(\phi)$.

3.1. Canonical measure. The analogue of the differential of a function is the map $d : \Omega^0(G) \rightarrow \Omega^1(G)$,

$$df(a, b) = f(b) - f(a).$$

For $x \in V$ define $\delta_x \in \Omega^0(G)$ that takes the value 1 at x and 0 at other vertices, and for $e \in \vec{E}$ define $\delta_e \in \Omega^1(G)$ that takes the value 1 at e , -1 at the opposite of e and 0 at other edges. So, we may evaluate d at the basis $\{\delta_x\}_{x \in V}$ of $\Omega^0(G)$ to obtain $d\delta_x = -\sum_{e \in \vec{E}, \underline{e}=x} \delta_e$, where $\underline{(a, b)} = a$. This tells us that we can write d in terms of δ_e and δ_x as

$$d = -\sum_{x \in V} \sum_{e \in \vec{E}, \underline{e}=x} \delta_e \otimes \delta_x$$

by identifying $\Omega^0(G)$ with its dual. Its adjoint d^* would be given by permuting the terms

$$d^* = -\sum_{x \in V} \sum_{e \in \vec{E}, \underline{e}=x} \delta_x \otimes \delta_e$$

or, in a more explicit way, for $\omega \in \Omega^1(G)$,

$$d^*\omega(x) = - \sum_{e \in \vec{E}, e=x} \omega(e).$$

We define $\text{div} = -d^*$ so that (3.1) rewrites as $\text{div}\lambda = \phi$, yielding to the reformulation of the corresponding affine space

$$\mathcal{F}(\phi) = \{\lambda \in \Omega^1(G) : \text{div}\lambda = \phi\}.$$

Let us for now assume that G is connected. Remark first that the solution space may be empty since we always have, by antisymmetry, $\sum_{v \in V} \text{div}\lambda(v) = 0$ which is reminiscent of Stokes' theorem. But this is the only restriction as explained in Proposition 3.1. If $\lambda_0 \in \mathcal{F}(\phi)$ we have $\mathcal{F}(\phi) = \lambda_0 + \mathcal{F}(0)$ and $\mathcal{F}(0)$ is precisely $\text{Ker}(\text{div})$. So, if $\mathcal{F}(\phi)$ is non-empty, we can calculate

$$\dim(\mathcal{F}(\phi)) = \dim(\text{Ker}(\text{div})) = |E| - \dim(\text{Im}(d)) = |E| - |V| + 1,$$

where we used that $\text{Ker}(d) = \{\text{constant functions}\}$ so that $\dim(\text{Im}(d)) = \dim(\Omega^0(G)) - 1 = |V| - 1$.

Proposition 3.1. *If G is connected, the set $\mathcal{F}(\phi)$ is not empty if and only if*

$$(3.2) \quad \sum_{v \in V} \phi(v) = 0.$$

Moreover, if $S \subset E$ and $\Omega^1(S)$ denotes the space of antisymmetric functions on $\vec{S} \subset \vec{E}$, the composition

$$\varphi_S : \mathcal{F}(\phi) \xrightarrow{\text{inclusion}} \Omega^1(G) \xrightarrow{\text{restriction}} \Omega^1(S)$$

is a bijection if and only if $\mathcal{F}(\phi) \neq \emptyset$ and $(V, E \setminus S)$ is a spanning tree. In case φ_S is a bijection, the matrices of φ_S and φ_S^{-1} in the bases $\{\hat{e}, e \in E\}, \{\hat{e}, e \in S\}$ have integer coefficients.

Proof of Proposition 3.1. We already know that the condition (3.2) is necessary. Let us show it is sufficient. Consider first the case where $G = (V, E)$ is a **tree**. To solve $\text{div}\lambda = \phi$ we look for the edges where λ is most easily determined by the boundary conditions. Let us denote by ∂V the set of leaves of G . Then, the divergence condition yields $\lambda(e) = \phi(x)$ if $x \in \partial V$ and e is the unique edge starting from x .

Next, denote by $\text{int } V = V \setminus \partial V$ the vertices of G of degree larger than one. Let us consider edges with an endpoint whose other edges connect only to leaves or, equivalently, the edges in $E \setminus \partial E$ adjacent to leaves of $V \setminus \partial V$. Take one such leaf $x \in \text{int } V$ and notice that, since x is not a leaf of G , the set $V_{1,x}$ of vertices in ∂V connected to x is non-empty. Let us reduce our task to finding a solution on $V \setminus V_{1,x}$ as follows. If e denotes the edge of $E \setminus \partial E$ starting at x , using the previous case of x being a leaf yields that a solution should satisfy

$$\phi(x) = \left(\sum_{y \in V_{1,x}} \lambda(x, y) \right) + \lambda(e) = - \left(\sum_{y \in V_{1,x}} \lambda(y, x) \right) + \lambda(e) = - \left(\sum_{y \in V_{1,x}} \phi(y) \right) + \lambda(e)$$

which determines λ at e as linear combinations of different values ϕ with integer coefficients. Now, we consider the graph restricted to $V \setminus V_{1,x}$, and define $\phi(x) = \left(\phi(x) + \sum_{y \in V_{1,x}} \phi(y) \right)$ and notice that (3.2) is satisfied for $V \setminus V_{1,x}$. We may proceed by induction until the tree consists solely of leaves in which case the condition (3.2) is precisely the equation $\text{div}\lambda = \phi$. By this procedure we have seen that, for a tree, in case there is a solution, it is unique, and its value at $e \in E$ is a linear combination with integer coefficients of the values $\phi(x)$ for $x \in V$.

For a **general graph** G we may take a spanning tree T of G and try to solve the equation for T . We may choose the values of λ arbitrarily at the edges that are not in T . More precisely, if S is the set of edges not belonging to T , we may consider any antisymmetric function $\tilde{\lambda} : \vec{S} \rightarrow \mathbb{R}$ and look for a solution $\lambda \in \mathcal{F}(\phi)$ satisfying $\lambda|_{\vec{S}} = \tilde{\lambda}$. To be able to forget the edges in S we change ϕ to $\phi_{\text{new}} : V \rightarrow \mathbb{R}$ given by

$$\phi_{\text{new}}(x) = \phi(x) - \sum_{e \in \vec{S}, e=x} \tilde{\lambda}(e)$$

so that if the divergence of λ in T at x is $\phi_{\text{new}}(x)$, the divergence of the extension by $\tilde{\lambda}$ of λ in G at x would be $\phi(x)$. Notice that the sum of $\phi_{\text{new}}(x)$ for $x \in V$ is the same as the sum of $\phi(x)$. This holds because each edge in S appears twice in the sum, once with each orientation, and thus the contribution to the sum cancels due to the antisymmetry. Then (3.2) holds for T and $(\phi_{\text{new}}; \gamma_{\text{new}})$,

and we may find a unique solution $\lambda \in \mathcal{F}(\phi_{new})$ on T such that $\lambda(x)$ is a linear combination with integer coefficients of the values of ϕ_{new} , and thus also of the values of ϕ and $\tilde{\lambda}$. We extend λ by $\tilde{\lambda}$ and remark that it is a solution in $\mathcal{F}(\phi)$.

On the other hand, if λ is a solution in $\mathcal{F}(\phi)$ its restriction to the directed edge set of T is a solution in $\mathcal{F}(\phi_{new})$ so that there is only one solution λ in $\mathcal{F}(\phi)$ satisfying $\lambda|_{\tilde{S}} = \tilde{\lambda}$. Notice that this already shows that φ_S is a bijection when $(V, E \setminus S)$ is a spanning tree of G .

It is clear that if φ_S is a bijection then $\mathcal{F}(\phi)$ is not empty because $\Omega^1(S)$ would be empty which is impossible. So, for the rest of the proof we may assume that $\mathcal{F}(\phi)$ is not empty. By taking any $\lambda_0 \in \mathcal{F}(\phi)$, using that $\mathcal{F}(\phi) = \mathcal{F}(0)$ and that the map $\mathcal{F}(\phi) \rightarrow \Omega^1(S)$ is a composition

$$\mathcal{F}(\phi) \xrightarrow{\text{translation by } -\lambda_0} \mathcal{F}(0) \xrightarrow{\varphi_S} \Omega^1(S) \xrightarrow{\text{translation by } \lambda_0|_{\tilde{S}}} \Omega^1(S),$$

we find that it is enough to prove the statements for $\phi = 0$. Now, let us show that if φ_S is a bijection then $(V, E \setminus S)$ is a spanning tree. If $(V, E \setminus S)$ had a cycle $(v_1, \dots, v_k, v_{k+1})$ with $v_{k+1} = v_1$ we could define $\lambda(v_i, v_{i+1}) = -\lambda(v_{i+1}, v_i) = 1$ for $i \in \{1, \dots, k\}$ and zero elsewhere. Such λ would belong to $\mathcal{F}(0)$ but its image in $\Omega^1(S)$ would be zero so that φ_S would not be injective. To show that $(V, E \setminus S)$ is connected we may notice that if it were a disconnected forest we could find $S' \subset S$ such that $(V, E \setminus S')$ is a spanning tree. But this would imply that the values at S' determine the solution and, thus, determine the values at $S \setminus S'$. Then φ_S could not be surjective. $\square$

Proposition 3.2. *Suppose that G is connected. If $(V, E \setminus S)$ is a spanning tree, then for all $\phi : V \rightarrow \mathbb{R}$ satisfying (3.2),*

$$(\varphi_S)_* \mathcal{L}eb_{\mathcal{F}(\phi)} = (\sqrt{\# \text{ Spanning trees of } G}) \mathcal{L}eb_{\Omega^1(S)}.$$

Proof. Now, assuming that $(V, E \setminus S)$ is a spanning tree, let us look for the constant $C > 0$ such that

$$(\varphi_S)_* \mathcal{L}eb_{\mathcal{F}(0)} = C \mathcal{L}eb_{\Omega^1(S)}$$

Denoting the dimension of $\mathcal{F}(0)$ by $k = |E| - |V| + 1$, we want to study the map $(\varphi_S)_*$ induced on k -forms. This is equivalent to looking at the pushforward map on k -vectors

$$(\varphi_S)_* : \Lambda^k \mathcal{F}(0) \rightarrow \Lambda^k \Omega^1(S),$$

taking the dual and inverting the resulting map. The constant $C > 0$ would be found by taking normalized vectors $w_1 \in \Lambda^k \mathcal{F}(0)$, $w_2 \in \Lambda^k \Omega^1(S)$ and solving

$$(\varphi_S)_* w_1 = \pm C^{-1} w_2$$

or, what is the same, taking $C^{-1} = |\langle (\varphi_S)_* w_1, w_2 \rangle|$. The vector w_2 can be explicitly obtained as $\wedge_{e \in S} \delta_e$, where we have chosen an orientation for each edge $e \in E$ and an order to perform the product. We recall that δ_e takes the value 1 at e with our chosen orientation, -1 if we reverse the orientation and 0 at all other edges. To explicitly construct w_1 is less obvious since we would be dealing with $\text{Ker}(\text{div})$. We could instead use its orthogonal complement $\text{Im}(d)$. By fixing a leaf v_0 of $(V, E \setminus S)$ we may consider $\{d\delta_v\}_{v \in V \setminus \{v_0\}}$, where we recall that δ_v is the function which is 1 at v and 0 elsewhere. Since $\sum_{v \in V \setminus \{v_0\}} d\delta_v + d\delta_{v_0} = d1 = 0$, this family generates $\text{Im}(d)$. Now, we obtain an element of $\Lambda^k \mathcal{F}(0; 0)$ by taking $*(\wedge_{v \in V \setminus \{v_0\}} d\delta_v)$, where $*$ denotes the Hodge star operator. We have not yet normalized this element nor shown it is non-zero but let us calculate, using that $\wedge_{e \in S} \delta_e = \pm *(\wedge_{e \in E \setminus S} \delta_e)$,

$$\langle *(\wedge_{v \in V \setminus \{v_0\}} d\delta_v), \wedge_{e \in S} \delta_e \rangle = \pm \langle \wedge_{v \in V \setminus \{v_0\}} d\delta_v, \wedge_{e \in E \setminus S} \delta_e \rangle = \pm \det \langle d\delta_v, \delta_e \rangle_{v \in V \setminus \{v_0\}, e \in E \setminus S}.$$

This determinant is a sum over all bijections $\sigma : E \setminus S \rightarrow V \setminus \{v_0\}$ of the product $(-1)^\sigma \prod_{e \in E \setminus S} \langle d\sigma(e), \delta_e \rangle$, where the sign $(-1)^\sigma$ is only defined up to an overall sign. Notice that $\langle d\delta_x, \delta_e \rangle = \pm 1$ if and only if x is an endpoint of e and, if not, $\langle d\delta_x, \delta_e \rangle = 0$. So, for a bijection to contribute, the unique edge adjacent to v_0 should correspond to the unique other endpoint v_1 of this edge. Then, for every other edge adjacent to v_1 we do not have a choice but to take the endpoint different from v_1 . If we continue in this way we get that there is only one bijection that contributes and therefore

$$\langle *(\wedge_{v \in V \setminus \{v_0\}} d\delta_v), \wedge_{e \in S} \delta_e \rangle = \pm 1.$$

This proves in particular that $*(\wedge_{v \in V \setminus \{v_0\}} d\delta_v)$ is not zero so the explicit formula for w_1 as the normalized $*(\wedge_{v \in V \setminus \{v_0\}} d\delta_v)$ works. If $(V, E \setminus S)$ is not a spanning tree of G we already know that φ_S is not a

bijection so that the previous inner product is zero. Using that $\{\wedge_{e \in S} \delta_e\}_{|S|=k}$ forms an orthonormal basis of $\Omega^1(G)$ we can calculate the norm as the sum of squares of inner products to obtain

$$\| * (\wedge_{v \in V \setminus \{v_0\}} d\delta_v) \|^2 = \# \text{ Spanning trees of } G.$$

This yields that

$$\langle (\varphi_S)_* w_1, w_2 \rangle = \pm \frac{1}{\sqrt{\# \text{ Spanning trees of } G}}$$

which implies the final statement of the proposition. $\square$

In the following, be set

$$(3.3) \quad d \text{ Vol} = \frac{1}{\sqrt{\# \text{ Spanning trees of } G}} \mathcal{L}eb_{\mathcal{F}(\phi)},$$

so that

$$(3.4) \quad (\varphi_S)_* d \text{ Vol} = \text{Leb}_{\Omega^1(S)}.$$

3.2. Boundary and sieving of graphs. For $G = (V, E)$ a finite graph, recall that we denote by ∂V (resp. $\text{int}(V)$) the set vertices of G of degree 1 (resp. degree larger than 1) and by ∂E the set of edges adjacent to ∂V . If $\phi \in \Omega^0(G)$, we denote by $\partial\phi$ the restriction of ϕ to ∂V . Likewise, we denote by $\partial\lambda$ the restriction of $\lambda \in \Omega^1(G)$ to $\{(a, b), \{a, b\} \in \partial E\}$. By the previous section, for $R \subset V$, $\mathcal{F}(\phi) \neq \emptyset$ if and only if

$$\sum_{v \in R} \phi(v) = - \sum_{v \in V \setminus R} \phi(v).$$

Definition 3.3 (Sieving of graphs). Let $r \geq 1$. Let $G = (V, E)$ be a finite graph (non necessarily connected) and $W, W' \subset \partial V$ with a bijection $g : W \rightarrow W'$. The sieving of G along (R_1, R_2) is the graph $G_{W*W'} = (\tilde{V}, \tilde{E})$ obtained as follows:

- $\tilde{V}$ is the quotient of V by the equivalence relation generated by $(v, g(v))$ for $v \in W$,
- $\tilde{E} = \pi(E)$, where $\pi : V \times V \rightarrow \tilde{V} \times \tilde{V}$ is the quotient map.

This construction informally amounts to merge v and $g(v)$ for $v \in W$ and considering the resulting edge structure inferred by E . Since $W, W' \subset \partial V$, $|\tilde{E}| = |E|$ as long as each connected component of G has a size at least 3 (which will always be the case in the present paper). There is moreover a canonical bijection between E and $\tilde{E}$. Let us closely look at the behavior of the equation (3.1) with respect to the sieving of graphs.

In the following, if $G = (V, E)$ is a finite graph, $V = S_1 \sqcup \dots \sqcup S_r$ is a partition of V and $\phi \in \Omega^0(G)$, we write by abuse of notation $(\phi|_{S_1}, \dots, \phi|_{S_r})$ instead of ϕ to detail the decomposition of ϕ along this partition. Moreover, we write $\mathcal{F}_G(\phi)$ instead of $\mathcal{F}(\phi)$ to emphasize that the equation (3.1) is considered in the graph G .

Proposition 3.4 (Product formula). *Let $G_1 = (V_1, E_1), G_2 = (V_2, E_2)$ be two connected finite graphs, such that $V_1 \cap V_2 = \emptyset$. Let $W \subset \partial V_1, W' \subset \partial V_2$ and let $g : W \rightarrow W'$ be a bijection. Set $\tilde{G} = (G_1 \cup G_2)_{W*W'}$ and let $\phi \in \Omega^0(\tilde{G})$ be such that $\sum_{v \in \tilde{V}} \phi(v) = 0$. Let $\lambda \in \Omega^1(\tilde{G})$ and for $w \in W$, let us denote by $x_w = \lambda(\tilde{e})$ where $\tilde{e}$ is the unique edge of $\tilde{G}$ starting from $\tilde{w}$ and ending on V_1 . Then,*

- (1) $\lambda \in \mathcal{F}_{\tilde{G}}(\phi)$ if and only if
 - (a) $\sum_{v \in V_1 \setminus W} \phi(v) + \sum_{w \in W} x_w = 0$,
 - (b) $\lambda|_{E_1} \in \mathcal{F}_{G_1}(\phi_1)$ where $\phi_1 = (\phi|_{V_1 \setminus W}, \phi_W)$ and where $\phi_W(w) = x_w$ for $w \in W$,
 - (c) $\lambda|_{E_2} \in \mathcal{F}_{G_2}(\phi_2)$ where $\phi_2 = (\phi|_{V_2 \setminus W'}, \phi'_W)$ and where $\phi'_W(w') = \phi(w') - \phi_W(g^{-1}(w'))$ for $w' \in W'$.
- (2) For $w_0 \in W$, $K_1 \subset \Omega^1(G_1)$ and $K_2 \subset \Omega^1(G_2)$,

$$\begin{aligned} \text{Vol}_{\tilde{G}}((K_1 \times K_2) \cap \mathcal{F}_{\tilde{G}}(\phi)) &= \int_{\mathbb{R}^{|W|-1}} \text{Vol}_{G_1} [K_1 \cap \mathcal{F}_{G_1}(\phi|_{V_1}, (x_w)_{w \in W \setminus W_0}, y(x))] \\ &\quad \cdot \text{Vol}_{G_2} [K_2 \cap \mathcal{F}_{G_2}(\phi|_{V_2}, (\phi(\tilde{w}) - x_w)_{w \in W \setminus \{w_0\}}, \phi(\tilde{w}) - y(w))] dx, \end{aligned}$$

where $y(w)$ is the unique solution to $\sum_{v \in V_1 \setminus W} \phi(v) + \sum_{w \in W \setminus \{w_0\}} x_w + y(w) = 0$.

Proof. (1) Recall that $\lambda \in \mathcal{F}_{\tilde{G}}(\phi)$ if and only if

$$\forall x \in \tilde{V} : \sum_{e, \tilde{e}=x} \lambda(e) = \phi(x) .$$

Let $\lambda \in \mathcal{F}_{\tilde{G}}(\phi)$ and let $x \in V_1$. If $x \in V_1 \setminus W$, then

$$\sum_{v \in V_1 : \{v, x\} \in E_1} \lambda_{|E_1}(v, x) = \sum_{v \in \tilde{V} : \{v, x\} \in \tilde{E}} \mathbb{1}_{\{v, x\} \in E_1} \lambda(v, x) = \sum_{v \in \tilde{V} : \{v, x\} \in \tilde{E}} \lambda(v, x) = \phi(x) .$$

If $w \in W$, then $\sum_{v \in V_1 : \{v, w\} \in E_1} \lambda_{|E_1}(v, w) = x_w$ as required, which gives that $\lambda_{|E_1} \in \mathcal{F}_{G_1}(\phi_{V_1 \setminus W}, \phi_W)$. Using the divergence condition for ϕ , one must have

$$\sum_{v \in V_1} \phi(v) + \sum_{w \in W} x_w = 0 .$$

Similarly, $\lambda_{|E_2} \in \mathcal{F}_{G_2}(\phi_{V_2 \setminus W'}, (y_w)_{w \in W'})$ with $y_w = \lambda(e)$, where e is the unique edge of $\tilde{E}$ starting from w and ending on V_2 . By using the divergence condition on $\tilde{w} = \{w, g(w)\}$ for $w \in W$, one has $\phi(\tilde{w}) = x_w + y_{g(w)}$, and thus $y_{g(w)} = \phi(\tilde{w}) - x_w$.

Reciprocally, one checks that if $\lambda \in \Omega^1(\tilde{G})$ is such that $\lambda_{|E_1} \in \mathcal{F}_{G_1}(\phi_{V_1 \setminus W}, (x_w)_{w \in W})$ and $\lambda_{|E_2} \in \mathcal{F}_{G_2}(\phi_{V_2 \setminus W'}, (\phi(\tilde{w}) - x_{g^{-1}(w)})_{w \in W})$, then $\lambda \in \mathcal{F}(\phi)$.

(2) Set $s + 1 = |W| = |W'|$ and write $W = \{w_0, \dots, w_s\}$. Let e_i be the edge of E_1 adjacent to w_i . Let T_i be a spanning tree of G_i for $i \in \{1, 2\}$. Since each edge $\tilde{w}, w \in W$ is bivalent in $\tilde{G}$, removing the edges e_j , $1 \leq j \leq s$ from $T_1 \cup T_2$ yields a spanning tree T of $\tilde{G}$. By Proposition 3.1 applied to $S = T^c$, the restriction map φ_S is a bijection from $\mathcal{F}_{\tilde{G}}(\phi)$ to $\Omega^1(S)$. Moreover, by Proposition 3.2, $(\varphi_S)_* d \text{Vol} = \mathcal{L}eb_{\Omega^1(S)}$. Let us write $S_i = E_i \setminus T_i$ and $R = \{e_1, \dots, e_s\}$. Then,

$$S = T^c = [E_1 \setminus T_1] \cup [E_2 \setminus T_2] \cup R = S_1 \cup S_2 \cup R .$$

Let $K_1 \subset \mathbb{R}^{E_1}$ and $K_2 \subset \mathbb{R}^{E_2}$. Then, for $(t_1, t_2, x) \in \mathbb{R}^{S_1} \times \mathbb{R}^{S_2} \times \mathbb{R}^{s-1}$,

$$\lambda = \lambda(t_1, t_2, x) = \varphi_S^{-1}(t_1, t_2, x) \in \mathcal{F}_{\tilde{G}}(\phi) .$$

By the previous statement, this is equivalent to the fact that $\lambda_{|E_1} \in \mathcal{F}_{G_1}(\phi_{|V_1 \setminus W}, (x_w)_{w \in R}, y)$ where y is the unique solution to $\sum_{v \in V_1 \setminus R} \phi(v) + \sum_{w \in R} x_w + y = 0$, and $\lambda_{|E_2} \in \mathcal{F}_{G_2}(\phi_{|V_2 \setminus W'}, (\phi(\tilde{w}) - x_w)_{w \in R}, \phi(w_0) - y)$. Let $\varphi_{S_1}^x$ be the projection from $\mathcal{F}_{G_1}(\phi_{|V_1 \setminus W}, (x_w)_{w \in R}, y)$ to $\Omega^1(S_1)$ and $\varphi_{S_2}^x$ be the projection from $\mathcal{F}_{G_2}(\phi_{|V_2 \setminus W'}, (\phi(\tilde{w}) - x_w)_{w \in R}, \phi(w_0) - y)$ to $\Omega^1(S_2)$. Since $E_i \setminus S_i$ is a spanning tree of G_i , each map $\varphi_{S_i}^x$ is bijective. Moreover, since then $\varphi_{S_1}^x \circ \lambda(t_1, t_2, x)$ is well-defined and equal to t_1 , we have

$$\lambda_{|E_1} = (\varphi_{S_1}^x)^{-1}(t_1) \text{ and, likewise, } \lambda_{|E_2} = (\varphi_{S_2}^x)^{-1}(t_2) .$$

Therefore, $\lambda(t_1, t_2, x) \in K_1 \times K_2 \cap \mathcal{F}(\phi)$ if and only if $(\varphi_{S_1}^x)^{-1}(t_1) \in K_1$ and $(\varphi_{S_2}^x)^{-1}(t_2) \in K_2$. Hence,

$$\begin{aligned} \text{Vol}(K_1 \times K_2 \cap \mathcal{F}(\phi)) &= \int_{\mathbb{R}^{S_1} \times \mathbb{R}^{S_2} \times \mathbb{R}^s} \mathbb{1}_{\lambda(t_1, t_2, x) \in K_1 \times K_2} dt_1 dt_2 dx \\ &= \int_{\mathbb{R}^s} \left(\int_{\mathbb{R}^{S_1} \times \mathbb{R}^{S_2}} \mathbb{1}_{(\varphi_{S_1}^x)^{-1}(t_1) \in K_1} \mathbb{1}_{(\varphi_{S_2}^x)^{-1}(t_2) \in K_2} dt_1 dt_2 \right) dx \\ &= \int_{\mathbb{R}^s} \left(\int_{\mathbb{R}^{S_1}} \mathbb{1}_{(\varphi_{S_1}^x)^{-1}(t_1) \in K_1} dt_1 \right) \cdot \left(\int_{\mathbb{R}^{S_2}} \mathbb{1}_{(\varphi_{S_2}^x)^{-1}(t_2) \in K_2} dt_2 \right) dx \\ &= \int_{\mathbb{R}^s} \text{Vol} [K_1 \cap \mathcal{F}_{G_1}(\phi_{|V_1 \setminus W}, (x_w)_{w \in R}, y)] \\ &\quad \cdot \text{Vol} [K_2 \cap \mathcal{F}(\phi_{|V_2 \setminus W'}, (\phi(\tilde{w}) - x_w)_{w \in R}, \phi(w_0) - y)] dx . \end{aligned}$$

□

Proposition 3.5 (Contraction formula). *Let $G = (V, E)$ be a connected finite graph, $W, W' \subset \partial V$ with $W \cap W' = \emptyset$ and $g : W \rightarrow W'$ a bijection and set $G_{W*W'} = (\tilde{V}, \tilde{E})$. Then, for $\phi \in \Omega^0(G_{W*W'})$ such that $\sum_{v \in \tilde{V}} \phi(v) = 0$ and $K \subset \Omega^1(\tilde{G})$,*

$$\text{Vol}(K \cap \mathcal{F}_{\tilde{G}}(\phi)) = \int_{\mathbb{R}^{|W|-1}} \text{Vol} [K \cap \mathcal{F}_G(\phi_{|V \setminus (W \cup W')}, (x_w)_{w \in W}, (\phi(\tilde{w}) - x_{g^{-1}(w')})_{w \in W'})] dx .$$

Proof. The proof is similar to the one of Proposition 3.4. $\square$

4. THE THREE-HOLED SPHERE

The goal of this section is to establish Theorem 2.6 in the case where $(g, p) = (0, 3)$, that is, for the three-holed sphere. Section 4.1 gives an injection of honeycombs into flows which is Proposition 4.3. The latter thus gives a parametrization of triangular honeycombs. In Section 4.2, we recall a combinatorial model from [FT24] called *dual hive*. Section 4.3 and Section 4.4 show that there is an linear bijection with integer coefficients between dual hives and triangular honeycombs. We finally prove Theorem 2.6 for the three-holed sphere in Section 4.5.

4.1. Parametrization of triangular honeycombs. The goal of this section is to view triangular honeycombs as flows on a graph with prescribed divergence. Proposition 3.2 then yields a volume form on this set of flows. Recall that for $d \geq 0$, $\mathcal{G}_d$ denotes the set of isomorphism classes of colored graphs with ordered boundary appearing in $\{G(h), h \in \text{HONEY}_{n,d}\}$. For a honeycomb $h \in \text{HONEY}^G$ and an edge $e \in E$, let us denote by $\ell^h(e)$ the type of e , defined in Definition 2.3, (2).

Lemma 4.1 (Boundary determine type and colors). *Let $G \in \mathcal{G}_d$ and let $h \in \text{HONEY}^G$. Then, the type $\ell^h : E \rightarrow \{0, 1, 2\}$ and color $c^h : E \rightarrow \{0, 1, 3\}$ are independent of h .*

Proof. Remark that such label and color maps can be defined similarly for any honeycomb $h = (\mathcal{E}, c)$ of $S \subset T$ such that any segment $e \in \mathcal{E}$ is contained in some ray $\{x + \mathbb{R}e^{2(\ell+1)i\pi/3}\}$ for some $\ell \in \{0, 1, 2\}$. Let us call such a honeycomb *admissible* and let us prove by induction on the number M of inner vertices (that is, the number of vertices of $G(h)$ which are in the interior of T) the following : *for any admissible honeycomb h , the induced label and color map on $G(h)$ only depends on the type and color map of the boundary edges and on the order on the boundary vertices.*

If $M = 1$, then all edges of $G[h]$ are boundary edges, and the assertion holds. Let $M > 1$ and let h be a honeycomb such that $G(h) = (V, E)$ has M inner vertices. Suppose that there is $v \in \text{int}(V)$ which is adjacent to two boundary edges $e_1 = \{v, v_1\}, e_2 = \{v, v_2\}$ and one non boundary edge e . Then, the type e is uniquely determined by the relation $\{\ell(e), \ell(e_1), \ell(e_2)\} = \{0, 1, 2\}$ and the value of $\ell(e_1)$ and $\ell(e_2)$. Next, since the cyclic order of the boundary vertices is given, by Definition 2.3 the color $c(e)$ is uniquely determined by $c(e_1)$ and $c(e_2)$. Hence, on $\tilde{G} = (V \setminus \{v_1, v_2\}, E \setminus \{e_1, e_2\})$ the type and color of the boundary edges is known, and by induction, the type and colors of all edges of $\tilde{G}$ only depends on the graph structure and their value on the boundary.

Suppose that all vertices of G are adjacent to at most one boundary edge. Let $(v_1, \dots, v_m)$, $m \geq 1$ be the boundary vertices in the cyclic order. By hypothesis, there exist $(w_1, \dots, w_m)$ such that $\partial E = \{e^i := \{v_i, w_i\}, 1 \leq i \leq m\}$ and $w_i \neq w_j$ when $i \neq j$. We claim that there exists $w_i, w_{i'}$ such that $\{w_i, w_{i'}\} \in E$ and $\{\ell(\{w_i, w_{i'}\}), \ell(e_i), \ell(e_{i'})\} = \{0, 1, 2\}$. Let $\tilde{G} = (V \setminus V_1, E \setminus \partial E)$ and $\tilde{h} = (\mathcal{E} \setminus \partial \mathcal{E}, c|_{\mathcal{E} \setminus \partial \mathcal{E}})$ where $\partial \mathcal{E}$ are the segments corresponding to edges of ∂E . Then, $t(\tilde{h}) = \bigcup_{e \in \mathcal{E} \setminus \partial \mathcal{E}} e \subset \overset{\circ}{T}$, and thus there exists a unique connected component K_0 in $T \setminus t(\tilde{h})$ which is adjacent to ∂T . Let L be a boundary component of K_0 in $\overset{\circ}{T}$. Then, L is a close polygonal line with vertices $\{z_1, \dots, z_p\}$ enumerated in the cyclic order. At each z_i , L has an angle α_i so that $\alpha_i = 4\pi/3$ if $z_i \in \{w_1, \dots, w_n\}$.

Since L is a close polygonal curve, there are at least two consecutive vertices z_i, z_{i+1} such that $\alpha_i = \alpha_{i+1} = 4\pi/3$, and thus $z_i = w_{j_i}$ and $z_{i+1} = w_{j_{i+1}}$ for some $1 \leq i_j \neq i_{j+1} \leq m$. Moreover, $[w_{j_i}, w_{j_{i+1}}]$ is a segment, and thus $\{w_{j_i}, w_{j_{i+1}}\} \in E$. Since the angle from $\{v_{j_i}, w_{j_i}\}$ (resp. $\{v_{j_{i+1}}, w_{j_{i+1}}\}$) to $\{w_{j_i}, w_{j_{i+1}}\}$ is $-2\pi/3$ (resp. $2\pi/3$),

$$\{\ell(\{w_{j_i}, w_{j_{i+1}}\}), \ell(\{v_{j_i}, w_{j_i}\}), \ell(\{v_{j_{i+1}}, w_{j_{i+1}}\})\} = \{0, 1, 2\}.$$

Let $w_i, w_{i'}$ be such that $e := \{w_i, w_{i'}\} \in E$, $e^1 := \{w_i, w_{i'}\}$ and $e^2 := \{v_i, w_i\}$ satisfy the angle condition $\{\ell(e^1), \ell(e^2), \ell(e^3)\} = \{0, 1, 2\}$. Then, $\ell(e^3)$ is determined by $\ell(e^1)$ and $\ell(e^2)$. Let f^1, f^2 be the third edge around w_i (resp. $w_{i'}$). Then, $\ell(f^i)$ is determined by $\ell(e^i)$ and $\ell(e^i)$ for $i \in \{1, 2\}$. By the color condition from Definition 2.1, $c(e) = 3$ if $c(e^1) \neq c(e^2)$ and otherwise $c(e) = c(e^1) = c(e^2)$. Then, $c(f^1)$ and $c(f^2)$ are uniquely determined by $\{c(e), c(e^2), c(e^2)\}$. Let $\hat{G} = (V \setminus \{v_i, v_{i'}\}, E \setminus \{e, e^1, e^2\})$. Then, $\hat{h} = (\mathcal{E} \setminus \{e, e^1, e^2\}, c|_{\mathcal{E} \setminus \{e, e^1, e^2\}})$ is a honeycomb such that $G(\hat{h}) = \hat{G}$ has $M - 1$ inner vertices,

and such that the type and color of the boundary edges are known. By induction, the type and color of all edges of $\hat{G}$, and thus of G are known. $\square$

Let us provide a description of honeycombs with structure graph G in terms of flows. Suppose that $G = (V, E) \in \mathcal{G}_d$. By the condition (1) Definition 2.3, G has only vertices of degree 1 or 3 and thus v has three adjacent edges e_ℓ , $\ell \in \{0, 1, 2\}$. Denote by $\text{int}(V)$ the set of vertices of degrees 3 and let $v \in \text{int}(V)$. By (2) of Definition 2.1, the angle between two successive edges at v is $2\pi/3$ and by (2) of Definition 2.3, each edge is oriented along $e^{2(\ell+1)i\pi/3}$ for some $\ell \in \{0, 1, 2\}$. Hence, there exists a sign $s(v) \in \{-1, +1\}$ such that, up to a relabeling, $e_\ell \subset \{v - s(v)e^{2(\ell+1)i\pi/3}\mathbb{R}_{\geq 0}\}$. For a univalent vertex $v \in \partial V$ connected to a unique trivalent vertex $v' \in \text{int}(V)$, we set $s(v) = -s(v')$.

Let $v, v' \in V$ be such that $e = \{v, v'\} \in E$. By the previous reasoning, there exist ℓ, ℓ' such that $e \subset \{v - s(v)e^{2(\ell+1)i\pi/3}\mathbb{R}_{\geq 0}\}$ and $e \subset \{v' - s(v')e^{2(\ell'+1)i\pi/3}\mathbb{R}_{\geq 0}\}$. Necessarily, $\ell = \ell'$ and $s(v) = -s(v')$, see Figure 5.

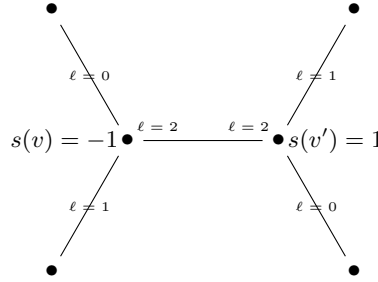


FIGURE 5. Two adjacent vertices in $\text{int}(V)$ for which $\ell = \ell' = 2$.

In particular, the map $s : v \mapsto s(v)$ only depends on the value of s on the boundary ∂V . Since $s(v)$ for $v \in \partial V$ is given by the type and thus the color of the unique adjacent edge, see Definition 2.3.(3), we deduce by Lemma 4.1 that s only depends on G . In the following definition, recall the definition of the height of an edge in Definition 2.3.

Definition 4.2 (Flow of a honeycomb). The *flow* of a honeycomb $h \in \text{HONEY}_{n,d}^G$ is the map $\mathcal{L}[h] \in \Omega^1(G)$ which to an oriented edge $(v, v') \in \vec{E}$ associates $\mathcal{L}[h](v, v') = s(v)L(\{v, v'\})$.

If $v \in \text{int}(V)$, v corresponds to a point of T and thus its coordinates (v_1, v_2, v_3) satisfies $v_1 + v_2 + v_3 = 1$. If e_1, e_2, e_3 are the three edges adjacent to v , we have by Definition 2.3(2), $L(e_1) + L(e_2) + L(e_3) = 1$. Hence, considering now oriented edges yields

$$(4.1) \quad \forall v \in \text{int}(V) : \sum_{v \sim v'} \mathcal{L}[h](v, v') = s(v) .$$

Let $(\alpha, \beta, \gamma) \in \mathcal{H}_{reg}^3$ and let $G \in \mathcal{G}_d$. Recall that for a structure graph G , its set of boundary edges $\partial E = \{\{v, v'\} \mid v \in \partial V\}$ consists of edges having an endpoint of degree one. Since by Definition 2.3, $\partial V \subset \partial T$, we may write $\partial E = (e_1, \dots, e_{3n})$, where for $0 \leq l \leq 2$ and $1 \leq i \leq n$, the edge $e_{\ell n+i}$ is adjacent to $v^{\ell n+i}$ on $\partial_\ell T$. Let us denote by $g^{\alpha, \beta, \gamma} \in \mathbb{R}^{\partial V}$ the boundary condition given by

$$g_{\ell n+i}^{\alpha, \beta, \gamma} := g^{\alpha, \beta, \gamma}(e_{\ell n+i}) = \begin{cases} \beta_i & \text{if } c(e_{\ell n+i}) = 0 \\ \beta_i - 1 & \text{if } c(e_{\ell n+i}) = 1 \end{cases}$$

for $\ell = 0$ and $1 \leq i \leq n$ and replacing β_i by α_i (resp. γ_i) when $\ell = 1$ (resp. $\ell = 2$). Remark that for $v \in \partial V$, $s(v) = 1$ if and only if $c(e) = 1$, where e is the unique edge adjacent to v , see for example Figure 2. Hence, the boundary condition translates into the condition

$$(4.2) \quad \mathcal{L}[h](v_{\ell n+i}, v') = g_{\ell n+i}^{\alpha, \beta, \gamma} ,$$

where v' is the unique vertex of V adjacent to $v_{\ell n+i}$. Hence, setting $\phi_{\alpha, \beta, \gamma}(v) = s(v)$ for $v \in \text{int}(V)$ and $\phi_{\alpha, \beta, \gamma}(v_{\ell n+i}) = g_{\ell n+i}^{\alpha, \beta, \gamma}$ for $v_{\ell n+i} \in \partial V$, (4.1) and (4.2) yields that for $h \in \text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)$,

$$\mathcal{L}[h] \in \mathcal{F}_G(\phi_{\alpha, \beta, \gamma}) .$$

Proposition 4.3 (Honeycomb injection). *Let $d \geq 0$ and $G \in \mathcal{G}_d$. The map $\mathcal{L} : h \rightarrow \mathcal{L}[h]$ is an injective map from HONEY^G to $\Omega^1(G)$ and*

$$\mathcal{L}(\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)) \subset \mathcal{F}_G(\phi_{\alpha, \beta, \gamma}) .$$

Proof. The fact that $\mathcal{L}(\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)) \subset \mathcal{F}_G(\phi_{\alpha, \beta, \gamma})$ is given by the previous discussion. Let us prove the injectivity of the map. Let $h_1, h_2 \in \text{HONEY}^G$ such that $\mathcal{L}[h_1] = \mathcal{L}[h_2]$. Since $G = (V, E)$ is isomorphic to the structure graph of both h_1 and h_2 , there are two isomorphisms $\iota_i : G \rightarrow (V^i, \mathcal{E}^i)$, $i \in \{1, 2\}$, where $\mathcal{E}^i$ is the set of segments associated to h_i by Definition 2.1 and V^i are the endpoints of these segments. Since both h and h' are honeycombs on the equilateral triangle, for which there exists a unique segment between two points, and $(V^1, \mathcal{E}^1)$ is isomorphic to $(V^2, \mathcal{E}^2)$, it suffices to show the equality $V^1 = V^2$.

Let $v \in V$. First, suppose that $v \in \partial V$. Since the boundary $\partial V \simeq (v^1, \dots, v^{3n})$ of G is ordered, there exists $1 \leq j \leq 3n$ such that $v = v^j$ and there exists a unique edge $e \in E$ adjacent to v . Since $(V^1, \mathcal{E}^1)$ and $(V^2, \mathcal{E}^2)$ are isomorphic to G as colored graphs with ordered boundary, $\iota_i(v)$ is the j -th vertex of the boundary of h_i and $c(\iota_i(e)) = c(e)$ for $i \in \{1, 2\}$. By Definition 2.3, $\iota_1(v)$ and $\iota_2(v)$ belong to the same boundary ∂_ℓ of T and their $(\ell + 1)$ -coordinates are

$$\begin{aligned} \iota_1(v)_{\ell+1} &= \delta_{c(\iota_1(e))=1} + (-1)^{\delta_{c(\iota_1(e))}=1} L(\iota_1(e)) = \delta_{c(e)=1} + (-1)^{\delta_{c(e)=1}} L(e) \\ &= \delta_{c(\iota_2(e))=1} + (-1)^{\delta_{c(\iota_2(e))}=1} L(\iota_2(e)) = \iota_2(v)_{\ell+1}. \end{aligned}$$

Hence, $\iota_1(v) = \iota_2(v)$.

Suppose that $v \in \text{int}(V)$. By Definition 2.3, v is a trivalent vertex and there are three edges e^0, e^1, e^2 adjacent to v . Moreover, by Lemma 4.1, the type and color of e^i is given by the graph structure and the type and color of the boundary edges. Suppose without loss of generality that for $1 \leq i \leq 2$, $\ell(e^i) = i$. Then,

$$\iota_1(v) = (L(e^0), L(e^1), L(e^2)) = \iota_2(v) .$$

We deduce that $V^1 = \iota_1(V) = \iota_2(V) = V^2$ and thus $h_1 = h_2$. □

4.2. Dual hive. Let us recall the definition of a *dual hive* from [FT24]. For $n \geq d \geq 0$, let us consider the graph $H_{d,n} = (R_{d,n}, E_{d,n})$ with vertices $R_{d,n}$ and edges $E_{d,n}$. Each vertex $v = r + se^{\pi i/3} \in H_{d,n}$ comes with a coordinate $(v_0, v_1, v_2) = (n+d-r-s, r, s)$. Each edge e of $H_{d,n}$ written $e = (v, v - e^{2\pi i \ell/3})$ with $\ell \in \{0, 1, 2\}$ is labelled $(\ell(e), h(e)) \in \{0, 1, 2\} \times \{0, \dots, n+d\}$ with

$$(4.3) \quad \ell(e) = \ell \text{ and } h(e) = v_\ell .$$

Table 1 below summarizes the different edge types for dual hives and honeycombs; edges of dual hives will be in duality with edges of honeycombs with the same type.

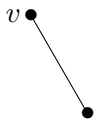
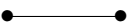
Type ℓ	$e = (v, v - e^{2\pi i \ell/3}) \in E_{n,d}$	$e \subset x + e^{2i\pi(\ell+1)/3} \in \text{HONEY}_{n,d}$
0		
1		
2		

TABLE 1. Edge types in dual hives and honeycombs.

Definition 4.4 (Color map). A *color map* is a map $C : E_{n,d} \rightarrow \{0, 1, 3, m\}$ such that the boundary colors around each triangular face in the clockwise order is either $(0, 0, 0)$, $(1, 1, 1)$, $(1, 0, 3)$ or $(0, 1, m)$ up to a cyclic rotation.

Definition 4.5 (Non-degenerate dual hive). For $(\alpha, \beta, \gamma) \in \mathcal{H}_{reg}^3$, such that $|\alpha| + |\beta| = |\gamma| + d$, the set of *dual hives*, denoted by $\text{DH}(\alpha, \beta, \gamma)$, is the set of pairs (C, L) such that :

- (1) $C : E_{n,d} \rightarrow \{0, 1, 3, m\}$ is a color map,
- (2) $L : E_{n,d} \rightarrow \mathbb{R}_{\geq 0}$ is the label map satisfying
 - (a) $L(e_1) + L(e_2) + L(e_3) = 1$ for every triangular face of $H_{d,n}$,
 - (b) if e, e' are edges of same type on the boundary of a same lozenge f ,
 - (i) $L(e) = L(e')$ if the middle edge of f is colored m ,
 - (ii) $L(e) > L(e')$ if $h(e) > h(e')$ and the middle edge of f is not colored m .
 - (c) The values of L on $\partial E_{n,d}$ are given by (α, β, γ) so that, sorted in decreasing height of edges, see Figure 6 below.

$$\begin{aligned} \ell^{(0,1)} &= (1 - \alpha_d, \dots, 1 - \alpha_1), & \ell^{(2,2)} &= (\alpha_{d+1}, \dots, \alpha_n) \\ \ell^{(2,0)} &= (1 - \beta_d, \dots, 1 - \beta_1), & \ell^{(1,1)} &= (\beta_n, \dots, \beta_{d+1}) \\ \ell^{(1,2)} &= (\gamma_n, \dots, \gamma_{n-d+1}), & \ell^{(2,2)} &= (1 - \gamma_{n-d}, \dots, 1 - \gamma_1). \end{aligned}$$

Moreover, for $\ell \in \{0, 1, 2\}$, the values of the color map C on $\partial^{(\ell, \ell)}$ is set to 0 while equal to 1 on other boundary edges. We call the triple (α, β, γ) the *boundary* of L , or of the dual hive.

Figure 7 shows an example of a dual hive for $d = 1$ and $n = 3$ with boundary

$$(4.4) \quad (\alpha, \beta, \gamma) = \left(\left(\frac{14}{23}, \frac{7}{23}, \frac{2}{23} \right), \left(\frac{18}{23}, \frac{10}{23}, \frac{3}{23} \right), \left(\frac{19}{23}, \frac{10}{23}, \frac{2}{23} \right) \right).$$

Colors red, blue, black and green correspond to values 0, 1, 3 and m of the color map respectively.

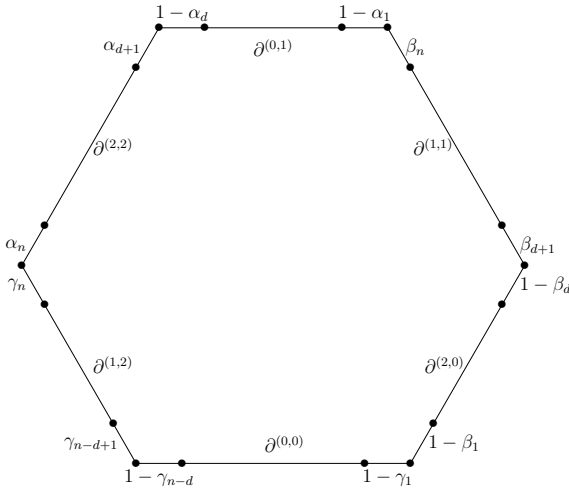


FIGURE 6. Boundary condition in $\text{DH}(\alpha, \beta, \gamma)$.

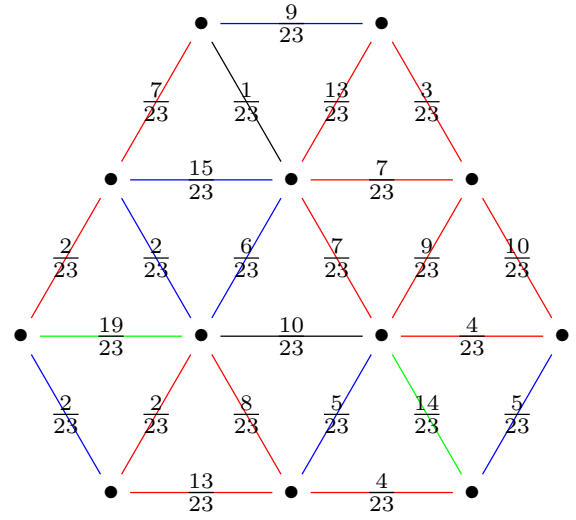


FIGURE 7. A dual hive with boundary condition (4.4).

For a given color map C , let us denote by $\text{DH}^C(\alpha, \beta, \gamma)$ the set of dual hives with boundary (α, β, γ) and color map C . Since an element of $\text{DH}^C(\alpha, \beta, \gamma)$ is uniquely defined by its map $L : E_{n,d} \rightarrow \mathbb{R}_{\geq 0}$, the set $\text{DH}^C(\alpha, \beta, \gamma)$ can be seen as an affine polytope of $\mathbb{R}^{E_{n,d}}$ written as

$$\text{DH}^C(\alpha, \beta, \gamma) = A^C \cap K_{n,d},$$

where $K_{n,d}$ is the cone of induced by (2)(b)(ii) and A^C is the affine subspace induced by the equalities coming from (2)(a), (2)(b)(i) and (2)(c).

4.3. From dual hive to triangular honeycomb.

Definition 4.6 ($\Gamma_{d,n}$ graph). Let $n \geq d$ be two integers. The *dual graph* $\Gamma_{d,n} = (V^\Gamma, E^\Gamma)$ of $H_{d,n}$ is the following graph :

- there is one vertex v_f for each triangular face f of $H_{d,n}$ and one vertex $v_{\tilde{e}}$ for each outer edge $\tilde{e}$ of $H_{d,n}$,
- there is an edge e between v_f and $v_{f'}$ (resp. between v_f and $v_{\tilde{e}}$) if the faces f and f' share an edge $\tilde{e}$ in $H_{d,n}$ (resp. if $\tilde{e}$ is a boundary edge of f in $H_{d,n}$).

The map $e \mapsto \tilde{e}$ yields a bijection from E^Γ to $E_{d,n}$ and e is then said *dual* to $\tilde{e}$. Hence, any color map $C : E_{d,n} \rightarrow \{0, 1, 3, m\}$ yields a color map, also denoted by C , from E^Γ to $\{0, 1, 3, m\}$ by setting $C(e) = C(\tilde{e})$. Likewise, any edge of E^Γ inherits the type $\ell(e) = \ell(\tilde{e}) \in \{0, 1, 2\}$, the height $h(e) = h(\tilde{e})$ and the label $L(e) = L(\tilde{e})$ of its dual edge.

To a dual hive $H = (C, L) \in \text{DH}(\alpha, \beta, \gamma)$, we associate a collection $\mathcal{S}(H)$ of segments of T :

- (1) For each $v \in V^\Gamma$:
 - if v is adjacent to three edges $(e^0, e^1, e^2) \in E_1^3$ with e^ℓ of type ℓ . We then set $x_v = (L(e^0), L(e^1), L(e^2)) \in T$,
 - if $v \in \partial V^\Gamma$ and v is adjacent to an edge e such that $\tilde{e} \in \ell^{(i,i)}$ (resp. in $\ell^{(i+1,i+2)}$), we set $x_v = (L(e)\delta_{i,0}, L(e)\delta_{i,1}, L(e)\delta_{i,2})$ (resp. $x_v = ((1-L(e))\delta_{i,0}, (1-L(e))\delta_{i,1}, (1-L(e))\delta_{i,2})$), where $\delta_{i,j} = 1$ if $i = j$ and 0 otherwise.
- (2) Then, we set

$$\mathcal{S}(H) = \{[x_v, x_{v'}] \mid e = \{v, v'\} \in E^\Gamma\}.$$

For $e = \{v, v'\} \in E^\Gamma$, let us denote its associated segment by

$$(4.5) \quad \Phi(e) = [x_v, x_{v'}].$$

As shown below, the collection of segments $\mathcal{S}(H)$ is almost the edge set of the structure graph of a honeycomb.

Lemma 4.7 (Edge segments). *Suppose that $\tilde{e} \in E_{n,d}$ is an edge of type ℓ adjacent to a face $\tilde{v}$ of $H_{n,d}$ and set $\epsilon = +1$ (resp. $\epsilon = -1$) if this face is a lower (resp. upper.) triangular face. Then, either $c(e) \neq m$ and*

$$\Phi(e) \subset x_v + \epsilon e^{2\pi i(\ell+1)/3} \mathbb{R}_{>0},$$

or $c(e) = m$ and

$$\Phi(e) = \{x_v\}.$$

Proof. Suppose without loss of generality that e is of type 0. Then, $(x_v)_0 = (x_{v'})_0 = L(e)$, so that $x_0 = L(e)$ for any $x \in \Phi(e) = [x_v, x_{v'}]$. We deduce that $\Phi(e) \subset v + \mathbb{R}e^{2\pi i/3}$.

If e is not colored m and is of the form $e = \{v_f, v_{f'}\}$, consider the lozenge of $H_{n,d}$ consisting of faces f and f' whose middle edge is $\tilde{e}$. Denote by $\tilde{f}, \tilde{f}'$ the two edges of type 2 of this lozenge, with the convention that $h(\tilde{f}') > h(\tilde{f})$ and $\tilde{f}$ (resp. $\tilde{f}'$) is a boundary edge of the face dual of v (resp. v'). Then, by Condition (2)(ii) of Definition 4.5, $L(\tilde{f}') > L(\tilde{f})$ and thus $(x_{v'})_2 > (x_v)_2$. If e is of the form $e = \{v_f, v_{\tilde{e}}\}$, we have that $x(v_{\tilde{e}})_2 = 0$ if $\tilde{e} \in \partial^{(0,0)}$ or $x(v_{\tilde{e}})_2 = 1 - L(e)$ if $\tilde{e} \in \partial^{(0,1)}$. The two previous cases correspond to $\epsilon = -1$ and $\epsilon = 1$ respectively. In both cases $\epsilon \cdot (x_{v_{\tilde{e}}})_2 > (x_v)_2$. We deduce that $e \subset x_v + \mathbb{R}_{>0}e^{2\pi i/3}$. A consequence of this fact is that the angle between two consecutive edges adjacent to an edge v is $2\pi/3$. If e is colored m Condition (2)(b)(i) of Definition 4.5 implies that $x_v = x_{v'}$ and thus $\Phi(e) = \{x_v\}$. $\square$

Lemma 4.8 (Distinct edges give disjoint segments). *If $e, e' \in E^\Gamma$ are distinct, then*

$$\text{int}(\Phi(e)) \cap \text{int}(\Phi(e')) = \emptyset.$$

This lemma is a rephrasing in the continuous case of the statements of Lemma [FT24, p. 5.9] and Lemma [FT24, p. 5.10]. We provide here a proof which is much simpler in its continuous version.

Proof. For $\tilde{e} \in E_{n,d}$, denote by $\tilde{e}_i = \tilde{v}_i$ (resp. $\tilde{e}^i$), where $\tilde{v}$ is the upper-triangular face (resp. lower-triangular) which is delimited by e . Let e, e' be of same type ℓ . First, by iterating Condition (2)(ii) of Definition 4.5, $L(e) > L(e')$ if $e_{\ell+1} = e'_{\ell+1}$ and $e_\ell > e'_\ell$. Next, using Condition (2)(ii) of Definition 4.5 and the fact that C is a color map, $L(e) > L(e')$ if $e_{\ell+1} = e'_{\ell+1} - 1$ and $e_\ell = e'_\ell + 1$. Therefore,

$L(e) > L(e')$ if e and e' are of same type ℓ and $e_{\ell+1} \leq e'_{\ell+1}$, $e_\ell > e'_\ell$. The same reasoning yields that $L(e) \geq L(e')$ if e and e' are of same type ℓ and $e_{\ell+1} \leq e'_{\ell+1}$, $e_\ell = e'_\ell$ with equality only if all edges of type $\ell - 1$ between e and e' are colored m .

Next suppose that $e = \{v_1, v_2\}$ and $e' = \{v'_1, v'_2\}$ with $e \neq e'$, with v_1, v'_1 being dual to an upper-triangular face and v_2, v'_2 being dual to a lower-triangular face. If $v_i = v'_j$ for some $i, j \in \{1, 2\}$, then $\text{int}(\Phi(e)) \cap \text{int}(\Phi(e')) = \emptyset$ by Lemma 4.7.

Otherwise, suppose without loss of generality that $(v_1)_0 < (v'_1)_0$. Since $\sum_{j=0}^2 (v_1)_j = \sum_{j=0}^2 (v'_1)_j = 1$, we can assume without loss of generality that $(v_1)_2 > (v'_1)_2$. Then, by the reasoning above, the edge e_1^2 (resp. e_2^2) of type 2 adjacent to v_1 (resp. v_2) satisfy $L(e_1^2) > L(e_2^2)$.

Set $x^i := x_{v_i}$ and $y^i = x_{v'_i}$ for $i = 1, 2$. Since $x_2^1 = L(e_1^2)$ and $y_2^1 = L(e_2^2)$, by the previous reasoning $x_2^1 > y_2^1$. Doing the same with the lower triangular faces v_2, v'_2 , which must be adjacent respectively to v_1 and v'_1 , yield that $x_2^2 \geq y_2^2$. Hence, the segments $\Phi(e) = [x^1, x^2]$ and $\Phi(e') = [y^1, y^2]$ can only meet at y^2 , and $\text{int}(\Phi(e)) \cap \text{int}(\Phi(e')) = \emptyset$. $\square$

Definition 4.9 (Maximal chain, reduced graph). Let $n \geq d$ be integers and let $C : E_{n,d} \rightarrow \{0, 1, 3, m\}$ be a color map.

- A *maximal chain* of C is a path $\gamma = (e_1, \dots, e_{2r+1}) \in (E_{n,d})^{2r+1}$ for some $r \geq 0$ such that $c(e_{2i}) = m$, $c(e_{2i+1}) = c(e_1)$ for $1 \leq i \leq r$, and such that two consecutive edges share a vertex. We write $\gamma = \{x, y\}$ for $x, y \in V^\Gamma$ where x (resp. y) is dual to a face f_x (resp. f_y) in $H_{n,d}$ such that $e_1 \in f_x$ (resp. $e_{2r+1} \in f_y$) and where f_x and f_y do not have any m edges to emphasize that the path goes from x to y . Moreover, the color of γ is defined as $c(\gamma) = c(e_1)$.
- The *reduced graph* of C is the graph $G^C = (V^C, E^C)$ defined by:
 - $V^C = \text{int}(V^C) \cup \partial(V^C)$, where $\text{int}(V^C) = V^\Gamma \setminus \{u \in V^\Gamma \mid \exists (u, v) \in E_\Gamma, C(\{u, v\}) = m\}$ and $\partial V^C = \partial V^\Gamma$,
 - $E^C = \{\gamma = \{x, y\} \mid \{x, y\} \text{ is a maximal chain of } C\}$.

The boundary vertices of G^C are ordered as the ones of Γ .

Remark that the definition of the edge set is valid, since any vertex $u \in V^\Gamma$ adjacent to an edge colored m cannot be the endpoint of a maximal chain of C . Moreover, by the color condition, a maximal chain with $c(e_1) = 3$ is necessarily of length 1. Note that any edge $e = \{x, y\} \in E_{n,d}$ not adjacent to an edge colored m is a maximal chain (with $r = 0$) and is thus in E^C .

The map $C \mapsto C^C$ is injective as we can recover C from G^C : it suffices to color the successive edges of a maximal chain γ as $c(e_{2i+1}) = c(\gamma)$ and $c(e_{2i}) = m$. In the sequel, we denote by ∂E^C the set of edges adjacent to a univalent vertex of G^C . Following Definition 4.5 and Definition 4.9, we introduce a partial order $\leq$ on E^C by completing the relation $e \leq e'$ if there exists an edge $\tilde{e} \in E_{d,n}$ (resp. $\tilde{e}' \in E_{d,n}$) dual to an edge in the equivalence class of e (resp. e') and such that $\tilde{e}, \tilde{e}'$ are of same type, adjacent to the same lozenge and satisfy $h(\tilde{e}') \geq h(\tilde{e})$.

Let C be a color map and let $H \in \text{DH}^C(\alpha, \beta, \gamma)$ be a dual hive. For any $\hat{e} \in E^C$, let us set $\hat{\Phi}[H](\hat{e}) = \bigcup_{e \in \hat{e}} \Phi[H](e)$ and

$$\rho_C(H) = \left(\{\hat{\Phi}[H](\hat{e}), \hat{e} \in E^C\}, c \right) = \bigcup_{e \in E} \Phi[H](e),$$

where $c : \{\hat{\Phi}[H](\hat{e}), \hat{e} \in E^C\} \rightarrow \{0, 1, 3\}$ with $c(\hat{\Phi}[H](\hat{e}))$ being the unique color different from m in the maximal chain $\hat{e}$. Remark that $\bigcup_{\hat{e} \in E^C} \hat{\Phi}[H](\hat{e}) = \bigcup_{e \in E} \Phi[H](e)$.

Lemma 4.10 (Reduced graph segments). *The set $\{\hat{\Phi}[H](\hat{e}) \mid \hat{e} \in E^C\}$ is a set of segments of T .*

Proof. Let $e = \{v_1, v_2\}$ and $e' = \{v'_1, v'_2\}$ be edges of E^Γ such that $\{v_2, v'_1\} \in E^\Gamma$ and $c(\{v_2, v'_1\}) = m$. Suppose without loss of generality that v_2 (resp. v'_1) is dual to an upper (resp. lower) triangular face of $H_{d,n}$. Let $\ell \in \{0, 1, 2\}$ be the type of edges e and e' . Then, by Lemma 4.7, $x_{v_2} = x_{v'_1} := x_v$, $\Phi(e) \subset x_{v_2} - \mathbb{R}e^{2(\ell+1)\pi/3}$ (resp. $\Phi(e') \subset x_{v_2} + \mathbb{R}e^{2(\ell+1)\pi/3}$) and $x_v \in \Phi(e) \cap \Phi(e')$, so that $\Phi(e) \cup \Phi(e')$ is a segment corresponding to $[x_{v_1}, x_{v'_2}]$. Hence, if $\hat{e} = \{v, v'\}$ is a maximal chain of C , $\bigcup_{e \in \hat{e}} \Phi(e)$ is the segment $[x_v, x_{v'}]$. $\square$

For $G \in \mathcal{G}_d$, recall that $\phi_{\alpha,\beta,\gamma} \in \Omega^0(V)$ has been defined before Proposition 4.3 in Section 4.1 and that the map $\mathcal{L} : \text{HONEY}_{n,d}^G \rightarrow \Omega^1(G)$ has been defined in Definition 4.2.

Proposition 4.11 (Dual hives as honeycombs). *Let $C : E_{n,d} \rightarrow \{0, 1, 3\}$ be a color map. The map ρ_C is a injection from $\text{DH}^C(\alpha, \beta, \gamma)$ to $\text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma)$ such that the map $\mathcal{L} \circ \rho_C : \text{DH}^C(\alpha, \beta, \gamma) \rightarrow \Omega^1(G^C)$ is the restriction of an affine map with integer coefficients from $\mathbb{R}^{E_{n,d}}$ to $\mathcal{F}_{G^C}(\phi_{\alpha,\beta,\gamma})$.*

Proof. Let us first prove that for $H \in \text{DH}^C(\alpha, \beta, \gamma)$, $\rho_C(H)$ is a triangular honeycomb. We first check that the two conditions of Definition 2.1 are fulfilled.

- (1) Suppose that $\hat{e} \neq \hat{e}'$ and $\text{int}(\hat{\Phi}(\hat{e})) \cap \text{int}(\hat{\Phi}(\hat{e}')) \neq \emptyset$. Let $x \in \text{int}(\hat{\Phi}(\hat{e})) \cap \text{int}(\hat{\Phi}(\hat{e}'))$. Since $\hat{\Phi}(\hat{e}) = \bigcup_{e \in \hat{e}} \Phi(e)$, $\hat{\Phi}(\hat{e}') = \bigcup_{e \in \hat{e}'} \Phi(e)$ and, by Lemma 4.8, $\text{int}(\Phi(e)) \cap \text{int}(\Phi(e')) = \emptyset$ for $e \neq e'$, we have that $x = x_v$ for some $v \in e \cap e'$ with $e \in \hat{e}$, $e' \in \hat{e}'$ not colored m . By Lemma 4.7 and up to switching e and e' , the angle from $\Phi(e)$ to $\Phi(e')$ is $2\pi/3$. Since $x \in \text{int}(\hat{\Phi}(\hat{e}))$, v is adjacent to a third edge colored m ; since C is a color map, $c(e) = 1$ and $c(e') = 0$.
- (2) Suppose that $x \in \partial\hat{\Phi}(\hat{e}) \cap \partial\hat{\Phi}(\hat{e}')$. Then, there exists $e \in \hat{e}$, $e' \in \hat{e}'$, neither of them colored m , such that $x \in \partial\Phi(e) \cap \partial\Phi(e')$. Then, Lemma 4.7 and the fact that C is a color map yields the second condition.

Hence, $\rho_C(H)$ is a honeycomb and the structure graph is given by

$$G[\rho_C(H)] = \left(\{x_v, v \in V^C\}, \left\{ \partial\hat{\Phi}[H](\hat{e}), \hat{e} \in G^C \right\} \right),$$

so that $G[\rho_C(H)]$ is isomorphic to G^C as colored graph with ordered boundary. We next turn to the conditions of being a triangular honeycomb.

- (1) Let x be a vertex of $G[\rho_C(H)]$. Then, x is the endpoint of a segment $\hat{\Phi}(\hat{e}) = \bigcup_{e \in \hat{e}} \Phi(e)$. Hence, $x = x_v$ for some $v \in V^\Gamma$ which is either dual to a triangular face $\tilde{v}$ without edge m on its boundary (for otherwise $x_v \in \text{int}(\hat{\Phi}(\hat{e}))$), or is equal to $v_{\tilde{e}}$ for some $\tilde{e} \in E_{d,n}$. In the first case, x is trivalent and, by Lemma 4.7, there are three non-trivial segments in T adjacent to x , with the angle between two successive segments being equal to $2\pi/3$: this implies that $x \in T \setminus \partial T$. In the second case, x is univalent and belongs to ∂T by construction.
- (2) Condition (2) is a direct consequence of Lemma 4.7.
- (3) Let x be the i -th boundary point of $\rho_C(H)$ on $\partial_1 T$, so that $x_0 = 0$. If $i \leq d$, then $x = x_{v_{\tilde{e}}}$ for the edge $\tilde{e} \in \partial^{(2,0)}$ such that $L(\tilde{e}) = 1 - \beta_i$ and $c(e) = 1$. Since $\tilde{e}$ is of type 2, $x_2 = L(\tilde{e}) = 1 - \beta_i$, and thus $x_1 = 1 - (1 - \beta_i) = \beta_i$. If $d + 1 \leq i \leq n$, then $\tilde{e} \in \partial^{(1,1)}$, $L(\tilde{e}) = \beta_i$ and $c(e) = 0$. Moreover, $\tilde{e}$ is of type 1 and thus $x_1 = L(\tilde{e}) = \beta_i$. The cases of other boundaries are similar.

Therefore, $\rho_C(H) \in \text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma)$.

Let us now check that $\mathcal{L} \circ \rho_C : \text{DH}^C(\alpha, \beta, \gamma) \rightarrow \Omega^1(G^C)$ is the restriction of an affine map with integer coefficients. Let $\hat{e} = \{v, v'\} \in E^C$ of type ℓ with $s(v) = 1$ and $s(v') = -1$, and suppose without loss of generality that $v_\ell < v'_\ell$. Let $\mathcal{E}(\hat{e}) = \{v, w\}$ be the unique edge of the maximal chain $\hat{e}$ adjacent to v . Then, $\mathcal{E}(\hat{e})$ is of type ℓ and $\mathcal{L}[\rho_C[H]](\hat{\Phi}[H](\hat{e})) = (x_v)_\ell = L(\mathcal{E}(\hat{e}))$. Hence, $\mathcal{L} \circ \rho_C$ is the restriction of the linear map from $\mathbb{R}^{E_{n,d}}$ to $\Omega^1(G^C)$ mapping $(x(e))_{e \in E_{n,d}}$ to $\sum_{\hat{e} \in E^C} x(\mathcal{E}(\hat{e})) \delta_{\hat{e}}$, where for $e = \{v, v'\} \in E^C$ with $s(v) = 1$ and $s(v') = -1$, $\vec{e} = (v, v')$. Remark that this map has integer coefficients in the canonical bases of both vector spaces.

Finally, since $h \in \text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)$ is uniquely determined by $(L(e))_{e \in G}$ by Proposition 4.3, the injectivity of the map $\mathcal{L} \circ \rho_C$ will be implied by the injectivity of the map ρ_C . Suppose that $H_1, H_2 \in \text{DH}^C(\alpha, \beta, \gamma)$ are distinct and denote by L_1, L_2 there respective label maps. Then, there exists $e \in E_{n,d}$ such that $L_1(e) \neq L_2(e)$. Denote by ℓ the type of e and, up to using Condition (2)(a) of Definition 4.5 on a triangular face next to e , assume that $c(e) \neq m$. Let $\hat{e} = \{v, v'\}$ be the maximal chain containing e , with the condition that $v_\ell < v'_\ell$. Then, L_i is constant on all edges $e \in \hat{e}$ not colored m , so that $L_1(\mathcal{E}(\hat{e})) = L_1(e) \neq L_2(e) = L_2(\mathcal{E}(\hat{e}))$. Hence, $\mathcal{L} \circ \rho_C(H_1) \neq \mathcal{L} \circ \rho_C(H_2)$, and ρ_C is injective. $\square$

4.4. From triangular honeycomb to dual hive. Let $h \in \text{HONEY}_{d,n}(\alpha, \beta, \gamma)$ be a triangular honeycomb with graph structure $G = (V^G, E^G) \in \mathcal{G}_d$. We construct a graph $\tilde{G} = (\tilde{V}, \tilde{E})$ with color and label $(\tilde{c}, \tilde{L})$ and a map $\tilde{\Phi} : \tilde{E} \rightarrow \mathcal{P}(T)$ as follows :

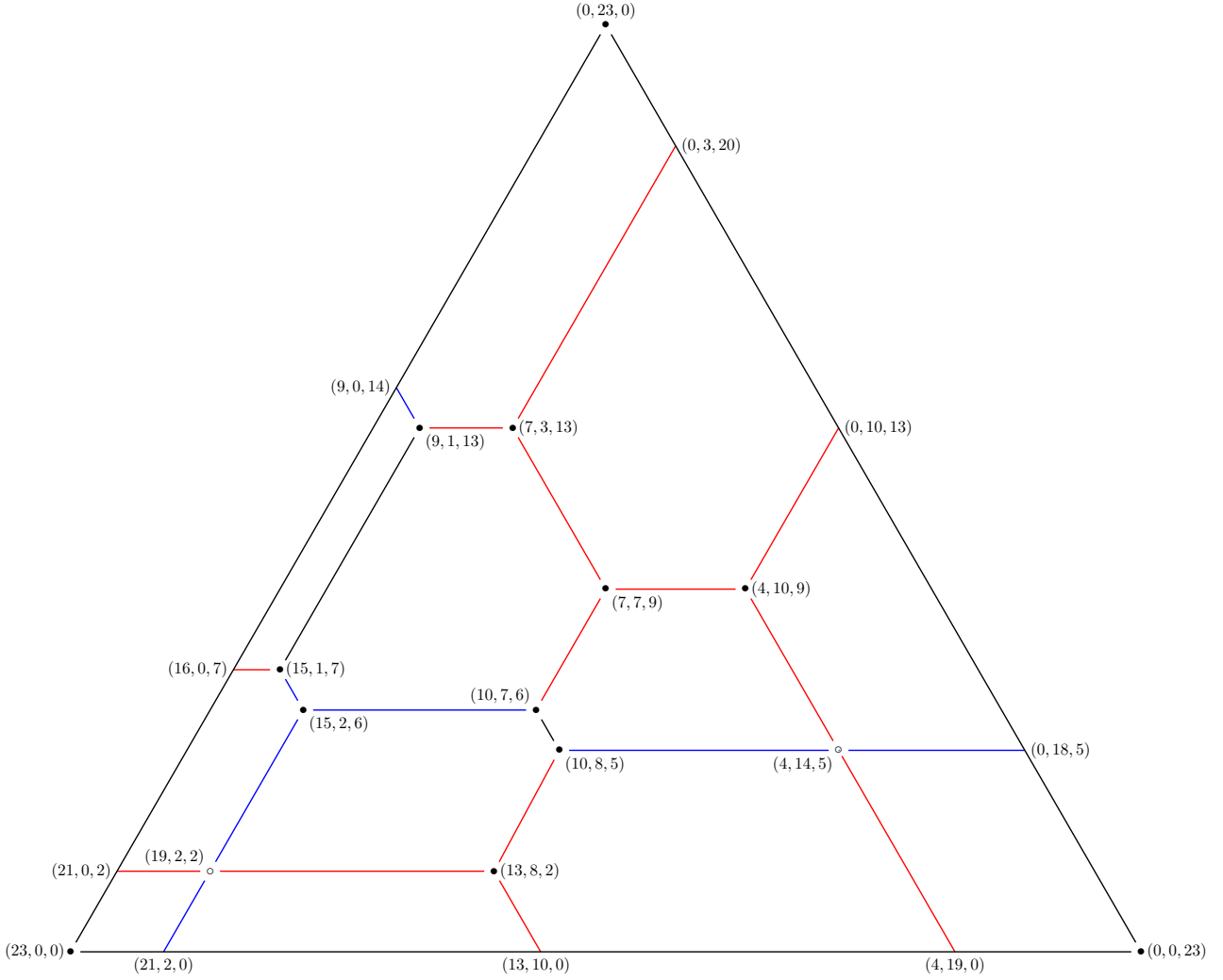


FIGURE 8. The triangular honeycomb corresponding to the dual hive of Figure 7. Coordinates should to be multiplied by $1/23$.

- (1) first consider an intermediate augmentation $\hat{G} = (\hat{V}, \hat{E})$ of G , where $\hat{V}$ consists of vertices V of G together with the points which are intersection of two segment of h . For $x, y \in \hat{V}$, we have an edge $\{x, y\} \in \hat{E}$ if and only if $[x, y] \subset e$ for some $e \in h$ and $]x, y[\cap \hat{V} = \emptyset$. Then, set $\tilde{c}(\{x, y\}) = c(e)$ and $\tilde{L}(\{x, y\}) = L(e) = x_{\ell(e)}$ if $]x, y[\subset e$. For $\{x, y\} \in \hat{E}$, one sets $\tilde{\Phi}(\{x, y\}) = [x, y] \in \mathcal{P}(T)$.
- (2) A vertex $v \in \hat{V}$ is of degree either 1 or 3 if it comes from a vertex of G or of degree 4 if it comes from a non-empty intersection $\iota(e) \cap \iota(e')$ for $e, e' \in E^G$. In the latter case, the four edges $\{\{v, x_i^\pm\}, i = 0, 1\}$ adjacent to v in $\hat{G}$ are such that $\{v, x_i^\pm\}$ is colored i and of type $\ell - i$ for some $\ell \in \{0, 1, 2\}$. In particular, the angle $\widehat{x_0^\epsilon v x_1^\epsilon} = 2\pi/3$ for $\epsilon \in \{-, +\}$. Replace v by two vertices v^+, v^- , add an edge e to $\hat{E}$ with color m , type $\ell + 1$ and label $1 - L(\{v, x_0^\pm\}) - L(\{v, x_1^\pm\})$ between v^+ and v^- . Set $\tilde{\Phi}(e) = \{v\}$. Replace each edge $\{v, x_i^\pm\}$ by $\{v^\pm, x_i^\pm\}$, keeping the same label and color. Repeat the operation successively for each vertex of degree 4.

The resulting augmentation $\tilde{G} = (\tilde{V}, \tilde{E})$ of G has univalent and trivalent vertices, each of the latter being adjacent to one edge of each type $\ell \in \{0, 1, 2\}$. Remark moreover that $\bigcup_{e \in \tilde{E}} e = \bigcup_{e \in E} e := \Lambda$. Hence, a connected region of $\mathbb{C} \setminus \Lambda$ is a polygon with angle either $2\pi/3$ or $\pi/3$. In the latter case, the vertex v of G is a vertex of degree 4 which has been replaced by two vertices of degree 3 and an edge in $\tilde{G}$ as in Figure 9. Hence, any bounded region of $\mathbb{C} \setminus \Lambda$ is bounded by 6 edges of $\tilde{G}$.

Therefore, the dual of $\tilde{G}$ is a graph $\tilde{H}$ with only triangular faces and inner vertices of degree 6. In particular, $\tilde{H}$ is isomorphic to a subgraph of the triangular grid. Let us define the type (resp. color,

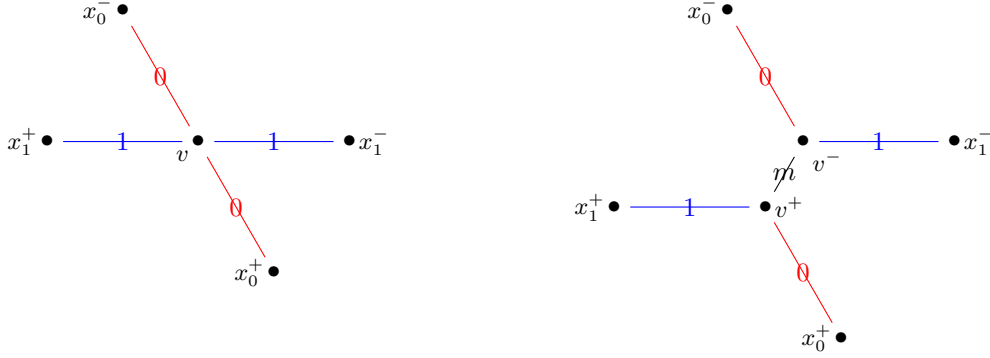


FIGURE 9. The augmentation of a vertex v of degree 4 to two vertices $v^\pm$ of degree 3 linked by an edge of type $\ell + 1 = 1$. The configuration is the one on the bottom right of Figure 8.

resp. label) of an edge e of $\tilde{H}$ as the same as the one his dual. In particular, each triangular face is bordered by three edges (e^0, e^1, e^2) , with e^ℓ of type ℓ and such that $L(e^0) + L(e^1) + L(e^2) = 1$. Since the types of the boundary edges is

$$(\underbrace{0, \dots, 0}_{n-d}, \underbrace{2, \dots, 2}_d, \underbrace{1, \dots, 1}_{n-d}, \underbrace{0, \dots, 0}_d, \underbrace{2, \dots, 2}_{n-d}, \underbrace{1, \dots, 1}_d),$$

$\tilde{H}$ is actually isomorphic to $H_{d,n}$. For $e \in E_{d,n}$, set $C(e) = \tilde{c}(\tilde{e})$ and $L(e) = \tilde{L}(\tilde{e})$

Lemma 4.12 (Honeycomb to dual hive). *The pair $H = (C, L)$ is a dual hive and $\rho_C(H) = h$.*

Proof. It is straightforward to check that the maps $C(e) = \tilde{c}(\tilde{e})$, $L(e) = \tilde{L}(\tilde{e})$ satisfy the properties (1), (2)(a), (2)(b)(i) and (c) of Definition 4.5.

To verify (2)(b)(ii), let $\tilde{e}, \tilde{e}'$ be opposite edges of type ℓ of a lozenge of $H_{d,n}$ with middle edge $\tilde{f}$ of type $\ell + 1$ not colored m and denote by e, e', f their dual edges in $E^{\tilde{G}}$. Suppose that $h(\tilde{e}) > h(\tilde{e}')$, where h has been defined in (4.3). We want to show that $L(\tilde{e}) = \tilde{L}(e) > L(\tilde{e}') = \tilde{L}(e')$. From the definition above, $\tilde{L}(e) = L(e)$ where for $e \in E^G$, $L(e) = x_\ell$ has been defined in Definition 2.3. We thus need to show that $x_\ell > x'_\ell$. Since $f \in E^{\tilde{G}}$ is of type $\ell + 1$, we have that $f \subset x + \mathbb{R}e^{2i\pi(\ell+2)/3}$ in T . The segment f cannot be reduced to a point by the definition of a non-degenerate honeycomb and for otherwise condition (2) of Definition 2.1 would not be satisfied. Therefore, the coordinate ℓ is strictly decreasing between edges e and e' so that $x_\ell > x'_\ell$. Hence, $H \in \text{DH}^C(\alpha, \beta, \gamma)$.

Remark that a triangular honeycomb $h = (\mathcal{E}, c)$ is uniquely determined by its image $t(h) = \bigcup_{e \in \mathcal{E}} e$, since then the elements of $\mathcal{E}$ are all the segments of $\mathcal{S}$ whose endpoints are univalent or trivalent vertices. Recall that for $e = \{v, v'\} \in E^\Gamma$, $\Phi(e) = [x_v, x_{v'}]$. Hence, to check that $\rho_C(H) = h$, it suffices to show that $\bigcup_{e \in E^\Gamma} \Phi(e) = \bigcup_{e \in h} e$. This is implied by the construction of $\tilde{\Phi}$ at the beginning of the section, since

$$\bigcup_{e \in E^\Gamma} \Phi(e) = \bigcup_{e \in \tilde{E}} \tilde{\Phi}(e) = \bigcup_{\{v, v'\} \in \tilde{E}} [x_v, x_{v'}] = \bigcup_{e \in h} e.$$

□

Putting together Lemma 4.11 with Lemma 4.12 yields the following decomposition.

Proposition 4.13 (Color map indexing). *There is a partition*

$$\text{HONEY}_{n,d} = \bigsqcup_{C \text{ color map}} \text{HONEY}_{n,d}^{G^C},$$

such that, for each color map C , the map ρ_C is a bijection and $(\mathcal{L} \circ \rho_C)^{-1}$ is the restriction of a linear map from $\mathcal{F}_{G^C}(\phi_{\alpha, \beta, \gamma})$ to $\mathbb{R}^{E_{n,d}}$ whose matrix in the canonical bases has integer coefficients.

Proof. By Lemma 4.11, the map $\rho_C : \text{DH}^C(\alpha, \beta, \gamma) \rightarrow \text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma)$ is injective.

Let $h \in \text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma)$ be a honeycomb. Then, by Lemma 4.12, there exists C' a color map and $H \in \text{DH}^{C'}(\alpha, \beta, \gamma)$ such that $\rho_{C'}(H) = h$. Hence, $h \in \text{HONEY}_{n,d}^{G^{C'}}(\alpha, \beta, \gamma)$. Since the map $C \mapsto G^C$

is injective, $C = C'$ and $H \in \text{DH}^C(\alpha, \beta, \gamma)$. Therefore, p_C is a bijection and $\text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma) \cap \text{HONEY}_{n,d}^{G^{C'}}(\alpha, \beta, \gamma) = \emptyset$ for $C \neq C'$. Therefore,

$$\text{HONEY}_{n,d} = \bigsqcup_{C \text{ color map}} \text{HONEY}_{n,d}^{G^C}.$$

Finally, for a color map C , the map ρ_C^{-1} is then obtained as follows : each segment $e \in h$ of type ℓ corresponds to a maximal chain $\hat{e}$ of type ℓ of $\Gamma_{n,d}$ with respect to C . Hence, for all edge $f \in \hat{e}$ of type ℓ , one has $\rho_C^{-1}[h](f) = L(e)$. Then, for any edge $f \in E_{n,d}$ colored m , one has $\rho_C^{-1}[h](f) = 1 - \rho_C^{-1}[h](f_1) - \rho_C^{-1}[h](f_2)$, where f_1 and f_2 are the two other edges of a triangular face bordered by f . Hence, the matrix of the map $(\mathcal{L} \circ \rho_C)^{-1}$ has integer coordinates in the canonical bases. $\square$

From the Proposition 4.13, any $G \in \mathcal{G}_d$ is of the form $G = G^C$ for some color map C . Since $V^C \subset V^\Gamma$, any vertex v of G^C inherits the coordinates of Γ by setting

$$(v_0, v_1, v_2) = (h(e_0), h(e_1), h(e_2)) ,$$

where e_ℓ is the edge of type ℓ adjacent to v in $\Gamma_{d,n}$ (even if $e_\ell \notin E^C$). For $e = \{v, w\}, e' = \{v', w'\} \in E^C$ of same type, introduce the cover relation $e < e'$ when, up to a transposition, $\{w, v'\}$ is an edge of E^C of type $\ell + 1$ and $v'_{\ell-1} > w_{\ell-1}$. This cover relation translates into a cover relation in $\vec{E}^C$ by saying that $(v, w) < (v', w')$ is and only if $\{v, w\} < \{v', w'\}$ in the former sense.

Corollary 4.14 (Parametrization of labels). *For a color map C and $G^C \in \mathcal{G}_d$,*

$$\mathcal{L} \left(\text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma) \right) = \mathcal{F}_{G^C}(\phi_{\alpha, \beta, \gamma}) \cap K_{G^C} ,$$

where $K_{G^C} \subset \Omega^1(G^C)$ is the cone defined as

$$K_{G^C} = \{ \omega \in \Omega^1(G^C) \mid |\omega(e)| < |\omega(e')| \text{ if } e < e' \} .$$

Proof. Let us define $\Psi : \Omega^1(G) \rightarrow \mathbb{R}^{E_{n,d}}$ by

$$\Psi[\omega](e) = \begin{cases} |\omega(\vec{e})| & \text{if } e \in \hat{e}, c(e) \neq m \\ 1 - |\omega(\vec{e}_1)| - |\omega(\vec{e}_2)| & \text{if } c(e) = m, e_1 \in \hat{e}_1, e_2 \in \hat{e}_2, (e_1, e_2, e) \text{ triangular face of } H_{n,d} \end{cases} .$$

By Proposition 4.13, we have that $\mathcal{L} \left(\text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma) \right) = \Psi^{-1}(\text{DH}^C(\alpha, \beta, \gamma))$.

Then, remark that $\text{DH}^C(\alpha, \beta, \gamma) = \mathcal{H}^C(\alpha, \beta, \gamma) \cap \mathcal{K}_<$, where $\mathcal{H}^C(\alpha, \beta, \gamma) \subset \mathbb{R}^{E_{n,d}}$ is the vector subspace determined by the conditions (2)(a), (2)(b)(i) and (2)(c) of Definition 4.5 and $\mathcal{K}_<$ is the cone given by

$$\mathcal{K}_< = \{ (H(e))_{e \in E_{n,d}} \mid L(e) < L(e') \text{ if } e, e' \text{ satisfy condition (2)(b)(ii) of Definition 4.5} \} .$$

Hence,

$$\mathcal{L} \left(\text{HONEY}_{n,d}^{G^C}(\alpha, \beta, \gamma) \right) = \Psi^{-1}(\mathcal{H}^C(\alpha, \beta, \gamma) \cap \mathcal{K}_\leq) = \Psi^{-1}(\mathcal{H}^C(\alpha, \beta, \gamma)) \cap \Psi^{-1}(\mathcal{K}_<) .$$

One then checks that $\Psi^{-1}(\mathcal{H}^C(\alpha, \beta, \gamma)) = \mathcal{F}_{G^C}(\alpha, \beta, \gamma)$ and $\Psi^{-1}(\mathcal{K}_<) = K_{G^C}$. $\square$

Remark 4.15 (Number of vertices and edges). By Proposition 4.13 and the construction of G^C from a coloured map C , we deduce that all the graphs G^C have the same number $n^2 + 3n$ of vertices and the same number $\frac{3n(n+1)}{2}$ of edges : indeed, vertices correspond to either triangles of the dual hive model which are not neighbours to an edge coloured m or to boundary edges, and edges correspond to edges of the dual hive model which are not coloured m . Since it has been proven in [FT24] that there are always $d(n-d)$ edges coloured m in a dual hive, the result is deduced.

4.5. Volume of flat connections. We can now combine the results of [FT24] with the ones of the previous section to prove Theorem 2.6 in the case of the three-holed sphere, that is, for $(g, p) = (0, 3)$. Let us denote by Σ_0^3 the Riemann sphere with three generic marked points removed. The moduli space of flat $\text{SU}(n)$ connections can be described as

$$M_{0,3}(\alpha, \beta, \gamma) = \{ (U_1, U_2, U_3) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma \mid U_1 U_2 U_3 = \text{Id}_{\text{SU}(n)} \} / \text{SU}(n) ,$$

where $SU(n)$ acts diagonally by conjugation on each factor. Its volume has been computed in [FT24] using two equivalent models named *toric hives* and *dual hives*. Using the results of this section, we present a reformulation of this results in terms of triangular honeycombs. For $G \in \mathcal{G}_d$, let us set

$$(4.6) \quad \text{Vol} [\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)] := \text{Vol} [\mathcal{L}(\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma))] ,$$

where $\mathcal{L} : \text{HONEY}_{n,d}^G(\alpha, \beta, \gamma) \rightarrow \mathcal{F}_G(\phi_{\alpha,\beta,\gamma})$ was defined in Definition 4.2 and Vol is the volume form defined in (3.3).

Theorem 4.16 (Volume of flat $U(n)$ -connections on the three-holed sphere). *Let $n \geq 3$ and consider the canonical volume form on $U(n)$. Then, for $\alpha, \beta, \gamma \in \mathcal{H}_{\text{reg}}$,*

$$Z_{0,3}(\alpha, \beta, \gamma) := \text{Vol} [M_{0,3}(\alpha, \beta, \gamma)] \neq 0$$

only if $\sum_{i=1}^n \alpha_i + \sum_{i=1}^n \beta_i + \sum_{i=1}^n \gamma_i = n + d$ for some $d \in \mathbb{N}$, in which case, if $\tilde{\gamma} = (1 - \gamma_n, \dots, 1 - \gamma_1)$,

$$Z_{0,3}(\alpha, \beta, \gamma) = \frac{2^{(n+1)[2]} (2\pi)^{(n-1)(n-2)}}{n! \Delta(\alpha) \Delta(\beta) \Delta(\gamma)} \sum_{G \in \mathcal{G}_d} \text{Vol} [\text{HONEY}_{n,d}^G(\alpha, \beta, \tilde{\gamma})] ,$$

where for $\alpha \in \mathcal{H}_{\text{reg}}$, $\Delta(\alpha) = 2^{n(n-1)/2} \prod_{i < j} \sin(\pi(\alpha_i - \alpha_j))$.

Before proving this theorem, let us recall three results from [FT24].

- (1) For any pair (C, C') of color maps, there exists a linear isomorphism $\text{Rot}[C \rightarrow C']$ from $\text{DH}^C(\alpha, \beta, \gamma)$ to $\text{DH}^{C'}(\alpha, \beta, \gamma)$ with integer coefficients on the canonical bases and such that $\text{Rot}[C \rightarrow C] = \text{Id}$ and $\text{Rot}[C \rightarrow C']^{-1} = \text{Rot}[C' \rightarrow C]$.
- (2) There exists a color map C_0 and a subset $S \subset E_{n,d}$ such that $p_S : \text{DH}^{C_0}(\alpha, \beta, \gamma) \rightarrow \mathbb{R}^S$ which to a label map $L : E_{n,d} \rightarrow \mathbb{R}$ associates $(L(e))_{e \in S}$ is an isomorphism such that both the map and its inverse have integer coefficients in the canonical bases indexed by $E_{n,d}$ and S .
- (3) We have the formula

$$Z_{0,3}(\alpha, \beta, \gamma) = \frac{2^{(n+1)[2]} (2\pi)^{(n-1)(n-2)}}{n! \Delta(\alpha) \Delta(\beta) \Delta(\gamma)} \sum_C \text{Vol}_S [\text{Rot}[C \rightarrow C_0] (\text{DH}^C(\alpha, \beta, \gamma))] ,$$

where the sum is over color maps $C : E_{n,d} \rightarrow \{0, 1, 3, m\}$ and where Vol_S is the Lebesgue measure of dimension $|S|$.

In the following proof, let us call an *integral* linear map $f : \mathbb{R}^E \rightarrow \mathbb{R}^F$ a linear map with integer coefficients in the canonical bases. Likewise, an integral isomorphism is an invertible linear map such that both a map and its inverse has integers coefficients in the canonical bases.

Proof of Theorem 4.16. Let C be a color map. By Proposition 3.1, there exists a set $R \subset E$ such the restriction map $\varphi_R : \mathcal{F}_{G^C}(\phi_{\alpha,\beta,\gamma}) \rightarrow \Omega^1(R)$ is an integral isomorphism. Denote by $i_R : \Omega^1(R) \rightarrow \Omega^1(G)$ the corresponding inverse map, which has thus affine with integer coefficients in the canonical basis and is a bijection from $\Omega^1(R)$ to $\mathcal{F}_{G^C}(\phi_{\alpha,\beta,\gamma})$. Since, by Proposition 4.13 and Corollary 4.14, $\Psi^C : \Omega^1(G^C) \rightarrow \mathbb{E}^{E_{n,d}}$ is an affine integral map and a bijection from $\mathcal{F}_{G^C}(\phi_{\alpha,\beta,\gamma}) \cap K_{G^C}$ to $\text{DH}^C(\alpha, \beta, \gamma)$ and by (1) above, $\text{Rot}[C \rightarrow C_0]$ is an integral isomorphism from $\text{DH}^C(\alpha, \beta, \gamma)$ to $\text{DH}^{C_0}(\alpha, \beta, \gamma)$. We deduce that

$$p_S \circ \text{Rot}[C \rightarrow C_0] \circ \Psi^C \circ i_R : \Omega^1(R) \rightarrow \mathbb{R}^S$$

is an integral isomorphism. Likewise, since by (2) above $p_S^{-1} : \mathbb{R}^S \rightarrow \text{DH}^{C_0}(\alpha, \beta, \gamma)$ is an integral affine isomorphism and $\mathcal{L} \circ \rho_C$ is an integral affine isomorphism from $\text{DH}^C(\alpha, \beta, \gamma)$ to $\mathcal{F}_{G^C}(\phi_{\alpha,\beta,\gamma})$,

$$F := (p_S \circ \text{Rot}[C \rightarrow C_0] \circ \Psi^C \circ i_R)^{-1} = \varphi_R \circ (\mathcal{L} \circ \rho_C) \circ \text{Rot}[C_0 \rightarrow C] \circ p_S^{-1} : \mathbb{R}^S \rightarrow \Omega^1(R)$$

is an integral isomorphism. We deduce that its determinant as an isomorphism from $\Omega^1(R)$ to $\mathbb{R}^S$ has modulus one. Hence,

$$\begin{aligned} \text{Vol}_S [\text{Rot}[C \rightarrow C_0] (\text{DH}^C(\alpha, \beta, \gamma))] &= \text{Leb} [u \in \mathbb{R}^S \mid \text{Rot}[C_0 \rightarrow C] \circ p_S^{-1}(u) \in \text{DH}^C(\alpha, \beta, \gamma)] \\ &= \text{Leb} [u \in \mathbb{R}^S \mid F(u) \in \varphi_R (\text{HONEY}_{n,d}^C(\alpha, \beta, \gamma))] \\ &= \text{Leb} [z \in \Omega^1(R) \mid z \in \varphi_R (\text{HONEY}_{n,d}^C(\alpha, \beta, \gamma))] \\ &= \text{Vol} [\text{HONEY}_{n,d}^C(\alpha, \beta, \gamma)] , \end{aligned}$$

where we used Proposition 3.2 on the last equality. Hence, by (3),

$$\begin{aligned} Z_{0,3}(\alpha, \beta, \gamma) &= \frac{2^{(n+1)[2]}(2\pi)^{(n-1)(n-2)}}{n!\Delta(\alpha)\Delta(\beta)\Delta(\gamma)} \sum_{C:E_{n,d} \rightarrow \{0,1,3,m\} \text{ color map}} \text{Vol}_S [\text{Rot}[C \rightarrow C_0] (\text{DH}^C(\alpha, \beta, \gamma))] \\ &= \frac{2^{(n+1)[2]}(2\pi)^{(n-1)(n-2)}}{n!\Delta(\alpha)\Delta(\beta)\Delta(\gamma)} \sum_{C:E_{n,d} \rightarrow \{0,1,3,m\} \text{ color map}} \text{Vol} [\text{HONEY}_{n,d}^{\Gamma^C}(\alpha, \beta, \gamma)] \\ &= \frac{2^{(n+1)[2]}(2\pi)^{(n-1)(n-2)}}{n!\Delta(\alpha)\Delta(\beta)\Delta(\gamma)} \sum_{G \in \mathcal{G}_d} \text{Vol} [\text{HONEY}_{n,d}^G(\alpha, \beta, \gamma)], \end{aligned}$$

where we used Proposition 4.13 for the last equality. $\square$

Remark 4.17 (Volume correspondance). Recall that in (2.2), we introduced the parametrization map

$$P : \text{HONEY}^G(\alpha, \beta, \gamma) \rightarrow \mathbb{R}^E,$$

which to a honeycomb h associates $P[h] = (|\mathcal{L}[h](e)|)_{e \in E}$. For $S \subset E$, let us consider the map

$$\begin{aligned} \eta_S : \Omega^1(S) &\rightarrow \mathbb{R}^S \\ w &\mapsto (|w(e)|)_{e \in S}. \end{aligned}$$

Then, the pullback $(\eta_S)^* \text{d Vol}$ of d Vol defined in (2.3) on $\mathbb{R}^S$ viewed as a subspace of $\mathbb{R}^E$ satisfies

$$(4.7) \quad (\eta_S)^* \text{d Vol} = \text{Leb}_{\Omega^1(S)} = (\varphi_S)^* \text{d Vol}',$$

where $\text{d Vol}'$ is the volume form defined in (3.3) and where we used (3.4) for the last equality.

5. SIEVING OF HONEYCOMBS

Let $\mathcal{T}$ be an oriented surface with boundary obtained by gluing N equilateral triangles $T^1, \dots, T^N$ by pairs along some of their boundaries in an orientation reversing way, such that p edges $L_1, \dots, L_p$ of the equilateral triangles are not glued together. For each $1 \leq i \leq N$, let $f_i : T \rightarrow T^i$ (viewed as a subset of $\mathcal{T}$) be an orientation-preserving isometry from T to T^i . For $1 \leq j \leq p$, let $1 \leq s_j \leq N$ and $0 \leq \ell_j \leq 2$ be such that $L_j = \partial_{\ell_j} T^{s_j}$. Each edge L_j has a natural orientation ℓ_j coming from the unique equilateral it belongs to.

Recall that a honeycomb on $\mathcal{T}$ is defined as a sequence $(h^i)_{1 \leq i \leq N}$ of triangular honeycombs such that $t_i(h^i)|_{\partial_j T^i} = t_{i'}(h^{i'})|_{\partial_{j'} T^{i'}}$ whenever $\partial_j T^i$ is glued to $\partial_{j'} T^{i'}$, where for a triangular honeycomb $h = (\mathcal{E}, c)$, $t_i(h) = \bigcup_{e \in \mathcal{E}} f_i(e)$.

Let $h = (h^i)_{1 \leq i \leq N}$ be such a honeycomb and denote by $G_i = (V_i, E_i)$ the structure graph of h^i for $1 \leq i \leq N$. Then, the structure graph $G(h)$ of h , as defined after Definition 2.4, can be describe as

$$G(h) = \left(\bigcup_{i=1}^N G_i \right)_{S * S'}$$

in the sense of Definition 3.3, where $S, S' \subset \bigcup_{i=1}^m \partial V_i$ are the vertices which are identified together. Hence, if $\mathcal{G}_{\mathcal{T}}$ denotes the set of structure graphs appearing in $\text{HONEY}_{\mathcal{T}}$, there is an injective map

$$i : \mathcal{G}_{\mathcal{T}} \rightarrow \mathcal{G}^N,$$

where $\mathcal{G}$ is the set of structure graphs appearing in the case $N = 1$ and $i(G)$ is the tuple $(G_1, \dots, G_N)$ such that $G = \left(\bigcup_{i=1}^N G_i \right)_{S * S'}$. Moreover, $G(h)$ has pn univalent vertices, n of them being on each boundary component L_j , $1 \leq j \leq p$, and $\frac{(3N-p)n}{2}$ bivalent vertices, n of them being on a same boundary of a triangle while being in the interior of the surface. For $1 \leq j \leq p$, let us denote by $(v_m^j)_{1 \leq m \leq n}$ the univalent boundary vertices on L_j ranked decreasingly with respect to their $(\ell_j + 1)$ -coordinate.

In Section 4.1, we defined for each $G_i = (V_i, E_i) \in \mathcal{G}_{d_i}$ a map $s_i : V_i \rightarrow \{-1, 1\}$ such that for each edge $\{v, v'\} \in E_i$, $s_i(v)s_i(v') = -1$. Let us denote by $\text{int}_3(V)$ the set of vertices of degree 3 of G . By the construction of G in terms of the graphs G_i and the fact that all vertices of $\text{int}(V_i)$ have degree 3,

$$\text{int}_3(V) = \bigsqcup_{1 \leq i \leq N} \text{int}(V_i).$$

We can therefore extend the maps s_i defined on each G_i to a map $s : \text{int}_3(V) \rightarrow \{-1, 1\}$.

Definition 5.1 (Label map of a honeycomb). Let $G = (V, E) \in \mathcal{G}_T$ and $h = (h_i)_{1 \leq i \leq N} \in \text{HONEY}_T^G$, with $h_i = (\mathcal{E}_i, c_i)$ for $1 \leq i \leq N$. Let $\phi : G(h) \rightarrow G$ be the corresponding graph isomorphism. Then, the *label map* $\mathcal{L}[h]$ is the element of $\Omega^1(G)$ whose value at $(v, v') \in \vec{E}$ such that $v \in \text{int}_3(V)$ and $\{v, v'\} = \phi(\partial e)$ for $e \in \mathcal{E}_i$ is

$$\mathcal{L}[h](v, v') = s(v)L(e) ,$$

where L is the coordinate map defined in Definition 2.3. (2) for h_i .

Remark that each edge $e \in E$ of G is adjacent to at least one trivalent vertex, and if $e = \{v, v'\}$ with $v, v' \in \text{int}_3(V)$, then necessarily there exists $1 \leq i \leq N$ such that $v, v' \in V_i$. Therefore $s(v)s(v') = -1$, so that $\mathcal{L}[h] \in \Omega^1(G)$. Hence, the latter construction yields a well-defined map

$$\mathcal{L} : \text{HONEY}_T^G \rightarrow \Omega^1(G).$$

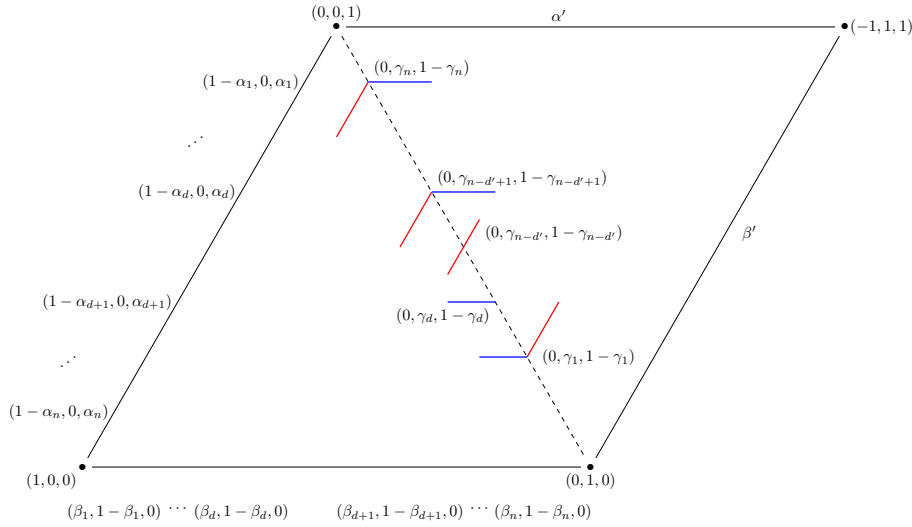


FIGURE 10. Line segments on the common edge of h and the rotation of h' in $\mathbb{R}_{\Sigma=1}^3$. Here, $d < d'$ which can be read from the orientation of line segments.

Let $\gamma^{(1)}, \dots, \gamma^{(p)} \in \mathcal{H}_{reg}$, and recall that $\text{HONEY}_T^G(\gamma^{(1)}, \dots, \gamma^{(p)})$ denotes the set of honeycombs h on $\mathcal{T}$ having boundary condition

$$(v_s^j)_{\ell_j} = \gamma_s^{(j)}.$$

Let us define $\phi_{\gamma^{(1)}, \dots, \gamma^{(p)}} : V \rightarrow \mathbb{R}$ by

$$(5.1) \quad \phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}(v) = \begin{cases} s(v) & \text{if } \deg v = 3 \\ 1 - c(v, v') - c(v, v'') & \text{if } \deg v = 2, v \sim v', v \sim v'', v' \neq v'' \\ \gamma_s^{(j)} & \text{if } v = v_s^j, v \sim v' \text{ and } c(v, v') = 0 \\ \gamma_s^{(j)} - 1 & \text{if } v = v_s^j, v \sim v' \text{ and } c(v, v') = 1 \end{cases}.$$

Lemma 5.2 (Flow of honeycomb). *For any $G \in \mathcal{G}_T$, $\mathcal{L}$ is injective and for $\gamma^{(1)}, \dots, \gamma^{(p)} \in \mathcal{H}_{reg}$,*

$$\mathcal{L}(\text{HONEY}_T^G(\gamma^{(1)}, \dots, \gamma^{(p)})) = \mathcal{F}_G(\phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}) \cap K_G ,$$

where $i(G) = (G_1, \dots, G_N)$, $\Omega^1(G)$ is identified with $\prod_{i=1}^N \Omega^1(G_i)$ and

$$K_G = K_{G_1} \times \dots \times K_{G_N},$$

with K_{G_i} defined in Corollary 4.14.

Let us introduce some notations to prove this lemma. Set $n(\mathcal{T}) = \frac{3N-p}{2}$ and denote by $B_1, \dots, B_{n(\mathcal{T})}$ the segments of $\mathcal{T}$ corresponding to boundary of triangles identified together. Each B_j is then adjacent to two triangles $T_j^{(1)}$ and $T_j^{(2)}$ and there exists $\ell_j^{(1)}, \ell_j^{(2)}$ such that $B_j = \partial_{\ell_j^{(1)}} T_j^{(1)} = \partial_{\ell_j^{(2)}} T_j^{(2)}$.

Proof. Let $G = (V, E) \in \mathcal{G}_T$. First, by definition,

$$\text{HONEY}_{\mathcal{T}}^G = \left\{ (h^1, \dots, h^N) \in \text{HONEY}_{T^1}^{G_1} \times \dots \times \text{HONEY}_{T^N}^{G_N} \mid h_{|B_j}^{r_j^{(1)}} = h_{|B_j}^{r_j^{(2)}}, 1 \leq j \leq n(\mathcal{T}) \right\}.$$

Next, recall that

$$V = \bigcup_{i=1}^N V_i / \left\langle \partial_{\ell_j} G^{r_j^{(1)}} = \partial_{\ell_j} G^{r_j^{(2)}}, 1 \leq j \leq n(\mathcal{T}) \right\rangle$$

and that E is the image of $\bigcup_{i=1}^N E_i$ in this quotient. Since any element of $\bigcup_{i=1}^N E_i$ has at most one endpoint in $\bigcup_{j=1}^{n(\mathcal{T})} \partial_{\ell_j} G^{r_j^{(1)}} \cup \partial_{\ell_j} G^{r_j^{(2)}}$, there is a canonical identification $\Omega^1(G) = \prod_{j=1}^N \Omega(G^j)$. Moreover, by the definition of $\mathcal{L} : \text{HONEY}_{\mathcal{T}}^G \rightarrow \Omega^1(G)$ from Definition 5.1, for $h \in \text{HONEY}_{\mathcal{T}}^G$

$$\mathcal{L}[h] = \left(\prod_{i=1}^N \mathcal{L} \right) (h^1, \dots, h^N)$$

under the previous identification. The injectivity of the map $\mathcal{L}$ from Proposition 4.3 yield the injectivity of $\mathcal{L}$.

It remains to describe the image $\mathcal{L}(\text{HONEY}_{\mathcal{T}}^G(\gamma^{(1)}, \dots, \gamma^{(p)}))$. By the previous reasoning, this amounts to describe the image through $\prod_{i=1}^N \mathcal{L}$ of the set

$$\left\{ (h^1, \dots, h^N) \in \text{HONEY}_{T^1}^{G_1} \times \dots \times \text{HONEY}_{T^N}^{G_N} \mid h_{|B_j}^{r_j^{(1)}} = h_{|B_j}^{r_j^{(2)}}, 1 \leq j \leq n(\mathcal{T}) \right\}.$$

Let us first describe how the condition $\partial_{\ell_j} h^{s_j} = \gamma^{(j)}$ for $1 \leq j \leq p$ translates through $\mathcal{L}$. Let $v_m^j \in \partial_{\ell_j} G_{s_j}$ and $x \in \partial_{\ell_j} h^{s_j}$ the point such that $\iota(v_m^j) = x$. Since $h^{s_j} \in \text{HONEY}_{G_{s_j}}$, like in Section 4.1, the condition $x_{\ell_j} = \gamma_m^{(j)}$ is equivalent to the condition

$$\mathcal{L}[h](v_m^j, v) = \gamma_m^{(j)} - c(v_m^j, v) = \phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}(v),$$

where v is the unique element of V_{s_j} such that $v \sim v_m^j$.

Let us now consider the condition $h_{|B_j}^{r_j^{(1)}} = h_{|B_j}^{r_j^{(2)}}$. Let $v \in \partial_{\ell_j^{(1)}} G_{r_j^{(1)}}$ and $v' \in \partial_{\ell_j^{(2)}} G_{r_j^{(2)}}$ such that $v \sim v'$ in G , and let $x = \iota_{T^{r_j^{(1)}}}(v)$, $x' = \iota_{T^{r_j^{(2)}}}(v')$ the corresponding image in $h_{|T^{r_j^{(1)}}}$ and $h_{|T^{r_j^{(2)}}}$. Using that the edges of both triangles are identified in the order-reversing way, the condition $h_{|B_j}^{r_j^{(1)}} = h_{|B_j}^{r_j^{(2)}}$ implies that

$$x_{\ell_j^{(1)}-1} = 1 - x'_{\ell_j^{(2)}-1}.$$

Let $w \in V_{r^{(1)}}$ and $w' \in V_{r^{(2)}}$ such that $v \sim_{G_{r^{(1)}}} w$ and $v' \sim_{G_{r^{(2)}}} w'$. Then, following (4.2), the previous equality is equivalent to $\mathcal{L}[h^{r^{(1)}}](v, w) + c(v, w) = 1 - \mathcal{L}[h^{r^{(1)}}](v, w) - c(v', w')$, which yields

$$\mathcal{L}[h](\bar{v}, w) + \mathcal{L}[h](\bar{v}, w') = 1 - c(v, w) - c(v', w') = \phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}(v).$$

□

In order to state the volume formula for (g, p) honeycombs, let us introduce for $r \in [0, 1)$ the notation

$$\mathcal{H}_{reg}^r = \left\{ \gamma \in \mathcal{H}_{reg} \mid \sum_{i=1}^n \gamma_i = r \pmod{\mathbb{Z}} \right\}.$$

This set is a union of affine polytopes of $\mathcal{H}_{reg}$ of dimension $n - 1$, and one can check that the volume du on each affine polytope induced by the projection on $\mathbb{R}^J$ is independent of J for any $J \subset \{1, \dots, n\}$ of cardinal $n - 1$. We simply denote by du this volume form.

Proposition 5.3. *Suppose that $\mathcal{T}'$ is obtained by gluing $\mathcal{T}$ and T^{N+1} along the boundaries L_p and $\partial_0 T^{N+1}$. Then, for all $G \in \mathcal{T}'$ and $\gamma^1, \dots, \gamma^{p+1} \in \mathcal{H}_{reg}$ such that $\sum_{i=1}^{p+1} |\gamma_i| \in \mathbb{N}$,*

$$\begin{aligned} & \sum_{G \in \mathcal{G}_{\mathcal{T}'}} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}'}^G(\gamma^1, \dots, \gamma^{p+1}) \right) \right] \\ &= \int_{\mathcal{H}_{reg}^\theta} \sum_{(G_1, G_2) \in \mathcal{G} \times \mathcal{G}_{\mathcal{T}}} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^{G_2}(\gamma^1, \dots, \gamma^{p-1}, u) \right) \right] \text{Vol} \left[\mathcal{L} \left(\text{HONEY}^{G_1}(\tilde{u}, \gamma^p, \gamma^{p+1}) \right) \right] du, \end{aligned}$$

where $\tilde{u} = (1 - u_n, \dots, 1 - u_1)$ and $\theta = -\sum_{i=1}^{p-1} |\gamma^i| \pmod{\mathbb{Z}}$.

Proof. Remark that from (5.1),

$$\sum_{v \in V} \phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}(v) = \sum_{j=1}^p \sum_{i=1}^n \gamma_s^{(j)} + \sum_{i=1}^N \left(\sum_{v \in \text{int}(V_i)} s(v) - \sum_{v \in \partial V_i, v \sim v'} c(v, v') \right).$$

Hence, in order for $\phi_{\gamma^{(1)}, \dots, \gamma^{(p)}}$ to yields a non-empty set $\mathcal{F}_G(\phi_{\gamma^{(1)}, \dots, \gamma^{(p)}})$, by Proposition 3.1 it is necessary that $\sum_{i=1}^p |\gamma^{(i)}| \in \mathbb{N}$. Hence, $\text{HONEY}_{\mathcal{T}}^{G_2}(\gamma^1, \dots, \gamma^{p-1}, u)$ is non-empty if and only if $u \in \mathcal{H}_{reg}^\theta$ with $\theta - \sum_{i=1}^{p-1} |\gamma^i| \pmod{\mathbb{Z}}$. Using Proposition 3.4 (2) gives

$$\begin{aligned} & \int_{\mathcal{H}_{reg}^\theta} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^{G_2}(\gamma^1, \dots, \gamma^{p-1}, u) \right) \right] \text{Vol} \left[\mathcal{L} \left(\text{HONEY}^{G_1}(\tilde{u}, \gamma^p, \gamma^{p+1}) \right) \right] du \\ &= \text{Vol} \left[(K_{G_1} \times K_{G_2}) \cap \mathcal{F}_{(G_1 \cup G_2)_{S_1 * S_2}} \left(\phi_{\gamma^{(1)}, \dots, \gamma^{(p+1)}} \right) \right]. \end{aligned}$$

By Lemma 5.2, the latter volume is exactly

$$\text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^G \left(\gamma^{(1)}, \dots, \gamma^{(p+1)} \right) \right) \right].$$

Therefore,

$$\begin{aligned} & \sum_{(G_1, G_2) \in \mathcal{G} \times \mathcal{G}_{\mathcal{T}}} \int_{\mathcal{H}_{reg}^\theta} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^{G_2}(\gamma^1, \dots, \gamma^{p-1}, u) \right) \right] \text{Vol} \left[\mathcal{L} \left(\text{HONEY}^{G_1}(\tilde{u}, \gamma^p, \gamma^{p+1}) \right) \right] du \\ &= \sum_{G \in \mathcal{G}_{\mathcal{T}'}} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^G \left(\gamma^{(1)}, \dots, \gamma^{(p+1)} \right) \right) \right]. \end{aligned}$$

□

A similar reasoning using Proposition 3.5 yields the following proposition.

Proposition 5.4. *Suppose that $\mathcal{T}'$ is obtained by gluing two boundaries of $\mathcal{T}$. Then, for each $G \in \mathcal{G}_{\mathcal{T}'}$, $\text{HONEY}_{\mathcal{T}'}^G$ admits a volume form with, for $\gamma^1, \dots, \gamma^{p-2} \in \mathcal{H}_{reg}$ with $\sum_{i=1}^{p-2} |\gamma^i| \in \mathbb{Z}$,*

$$\sum_{G \in \mathcal{G}_{\mathcal{T}'}} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}'}^G \left(\gamma^{(1)}, \dots, \gamma^{(p-2)} \right) \right) \right] = \int_{\mathcal{H}_{reg}} \sum_{G \in \mathcal{G}_{\mathcal{T}}} \text{Vol} \left[\mathcal{L} \left(\text{HONEY}_{\mathcal{T}}^G \left(\gamma^{(1)}, \dots, \gamma^{(p-2)}, u, \tilde{u} \right) \right) \right] du.$$

6. PROOF OF THEOREM 2.6 : VOLUME OF FLAT $U(n)$ -CONNECTIONS ON A COMPACT SURFACE

The goal of this section is the proof of Theorem 2.6, which gives a volume expression for the volume $M_{g,n}(\alpha_1, \dots, \alpha_p)$ of flat $SU(n)$ -connections on a surface $\mathcal{M}$ of genus g with p boundary components for $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{reg}$. Recall that $\mathcal{H}_{reg}^0$ denotes the set of regular conjugacy classes of $SU(n)$.

6.1. Parametrizations of conjugacy classes and volume form. Let us consider the standard parametrization of conjugacy classes in $SU(n)$ given by

$$\mathcal{A} = \left\{ t_1 \geq \dots \geq t_n \mid \sum_{i=1}^n t_i = 0, t_1 - t_n \leq 1 \right\}.$$

The set $\mathcal{A}$ is called an alcove of type A_{n-1} . Remark that $\mathcal{A}$ is a polytope of dimension $n-1$ in $\mathbb{R}^n$, and for any $R \subset \{1, \dots, n\}$ of cardinal $n-1$, the projection $p_R : \mathcal{A} \rightarrow \mathbb{R}^R$ yields a non-zero volume

form $p_R^* d\ell_{\mathbb{R}^R}$ on $\mathcal{A}$. This volume form is independent of R and denoted by dt in the sequel. Choosing for example $R = \{1, \dots, n-1\}$, we have

$$\begin{aligned} \text{Vol}(\mathcal{A}) &= \int_{\mathbb{R}^{n-1}} \mathbb{1}_{t_1 \geq \dots \geq t_{n-1}, t_1 + \sum_{i=1}^{n-1} t_i \leq 1} \prod_{i=1}^{n-1} dt_i \\ &= \int_{\mathbb{R}^{n-1}} \mathbb{1}_{1 \geq u_1 \geq \dots \geq u_{n-1} \geq 0} \frac{1}{n} \prod_{i=1}^{n-1} du_i = \frac{1}{n!}, \end{aligned}$$

where we did the change of variable $\eta : (t_i)_{1 \leq i \leq n-1} \mapsto (t_i + \sum_{j=1}^{n-1} t_j)_{1 \leq i \leq n-1}$ with $\text{Jac}(\eta(t)) = n$.

Denote by $\mathcal{A}_{\text{reg}}$ the subset of $\mathcal{A}$ consisting of tuples $(t_1 > \dots > t_n)$ with $t_1 - t_n < 1$. The set $\mathcal{A}_{\text{reg}}$ parametrizes then the regular conjugacy classes of $\text{SU}(n)$, as $\mathcal{H}_{\text{reg}}^0$ does. There is moreover a piecewise affine bijection $\phi : \mathcal{A}_{\text{reg}} \rightarrow \mathcal{H}_{\text{reg}}^0$ whose value on $(t_1 > \dots > t_i > 0 > t_{i+1} > \dots > t_n)$ is

$$(6.1) \quad \phi(t_1, \dots, t_n) = (1 + t_{i+1}, \dots, 1 + t_n, t_1, \dots, t_i).$$

One has $\text{Jac}(\phi(t)) = 1$ for all $t \in \mathcal{A}_{\text{reg}}$, and thus ϕ is volume preserving. One then checks that $\phi^* d\theta = dt$.

6.2. Contraction formula on moduli spaces of flat connections. Let $\widehat{\mathcal{M}}$ be a (possibly disconnected) oriented surface with $(p+2)$ boundary components $L_1, \dots, L_p, L_{p+1}, L_{p+2}$. Let $\mathcal{M}$ be the oriented surface obtained by gluing L_{p+1} and L_{p+2} in an orientation reversing way and suppose that $\mathcal{M}$ is connected. Then, if we denote by $M(\mathcal{M}, \alpha_1, \dots, \alpha_p)$ the moduli space of flat $\text{SU}(n)$ -connections on $\mathcal{M}$, the following formula from [MW99a] holds for its volume.

Theorem 6.1 ([MW99a, Prop. 5.4]). *Suppose that $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}^0$. If $M(\mathcal{M}, \alpha_1, \dots, \alpha_p)$ contains at least one connection whose stabilizer is $Z(\text{SU}(n))$, then*

$$\text{Vol}(M(\mathcal{M}, \alpha_1, \dots, \alpha_p)) = \frac{1}{k} \int_{\mathcal{A}} \text{Vol}(M(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \phi(t), \phi(-t))) \Delta(t)^2 dt,$$

where $k = 1$ if $\widehat{\mathcal{M}}$ is connected and $\#Z(\text{SU}(n)) = n$ otherwise, dt is the previous volume form on $\mathcal{A}$ and $\Delta(t) = 2^{n(n-1)/2} \prod_{1 \leq i < j \leq n} \sin(\pi(t_i - t_j))^2$.

Note that we slightly adapted the result of [MW99a], which was stated for the torsion volume of [Wit92], to translate it in the symplectic picture : this only consists in adding the term $\Delta(t)$. Let us first remark that considering $\text{U}(n)$ -valued connection instead of $\text{SU}(n)$ -valued connection does not change the volume. Indeed, suppose that $\mathcal{M}$ corresponds to a surface of genus g with p boundary components. Then, for any $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}$,

$$\begin{aligned} M_{\text{U}(n)}(\mathcal{M}, \alpha_1, \dots, \alpha_p) \\ \simeq \left\{ ((U_i)_{1 \leq i \leq 2g}, C_1, \dots, C_p) \in \text{U}(n)^{2g} \times \mathcal{O}_{\alpha_1} \times \dots \times \mathcal{O}_{\alpha_p} \mid \prod_{i=1}^g [U_{2i-1}, U_{2i}] = \prod_{i=1}^p C_i \right\} / \text{U}(n), \end{aligned}$$

where $\text{U}(n)$ acts diagonally by conjugation. Since $\det \prod_{i=1}^g [U_{2i-1}, U_{2i}] = 1$, the latter set is non-empty only if $\sum_{i=1}^p |\alpha_i| \in \mathbb{N}$. Next, remark that any conjugacy class of $\text{SU}(n)$ is also a conjugacy class of $\text{U}(n)$ and there is a natural action of $\mathbb{R}$ on $\mathcal{H}_{\text{reg}}$ given by

$$t \cdot (\theta_1 > \dots > \theta_n) = \text{std}(\theta_i + t \mod \mathbb{Z}),$$

where $\text{std}(x_1, \dots, x_n)$ denotes the standardization $(x_{i_1} > x_{i_2} > \dots > x_{i_n})$. For $\alpha \in \mathcal{H}_{\text{reg}}$, set $t(\alpha) = |\alpha| \mod \mathbb{Z}$ and $\hat{\alpha} = (-t_\alpha/n) \cdot \alpha$, so that $\hat{\alpha} \in \mathcal{H}_{\text{reg}}^0$. Since $Z(\text{U}(n))$ acts trivially by conjugation, when $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}$ we have

$$\begin{aligned} M_{\text{U}(n)}(\mathcal{M}, \alpha_1, \dots, \alpha_p) \\ \simeq \left\{ ((U_i)_{1 \leq i \leq 2g}, C_1, \dots, C_p) \in \text{U}(n)^{2g} \times \mathcal{O}_{\alpha_1} \times \dots \times \mathcal{O}_{\alpha_p} \mid \prod_{i=1}^g [U_{2i-1}, U_{2i}] = \prod_{i=1}^p C_i \right\} / \text{SU}(n) \\ \simeq \mathbb{T}^{2g} \times \left\{ ((U_i)_{1 \leq i \leq 2g}, C_1, \dots, C_p) \in \text{SU}(n)^{2g} \times \mathcal{O}_{\hat{\alpha}_1} \times \dots \times \mathcal{O}_{\hat{\alpha}_p} \mid \prod_{i=1}^g [U_{2i-1}, U_{2i}] = \prod_{i=1}^p C_i \right\} / \text{SU}(n) \\ \simeq \mathbb{T}^{2g} \times M_{\text{SU}(n)}(\mathcal{M}, \hat{\alpha}_1, \dots, \hat{\alpha}_p), \end{aligned}$$

so that with the convention that $\text{Vol}(\mathbb{T}) = 1$,

$$(6.2) \quad \text{Vol} [M_{\text{U}(n)}(\mathcal{M}, \alpha_1, \dots, \alpha_p)] = \text{Vol} [M_{\text{SU}(n)}(\mathcal{M}, \hat{\alpha}_1, \dots, \hat{\alpha}_p)] .$$

We then derive the following proposition. Recall that for $\theta = (\theta_1 > \dots > \theta_n) \in \mathcal{H}_{\text{reg}}$, we denote by $\tilde{\theta}$ the element of $\mathcal{H}_{\text{reg}}$ given by

$$\tilde{\theta} = (1 - \theta_n > \dots > 1 - \theta_1) .$$

Proposition 6.2. *Suppose that $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}^0$. If either $p \geq 3$ or $(p, g(\mathcal{M})) = (1, 1)$ or $g(\mathcal{M}) \geq 2$, then, if $\widehat{\mathcal{M}}$ is disconnected,*

$$\text{Vol}(M(\mathcal{M}, \alpha_1, \dots, \alpha_p)) = \frac{1}{n} \int_{\mathcal{H}_{\text{reg}}^0} \text{Vol}(M(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta ,$$

and, if $\widehat{\mathcal{M}}$ is connected,

$$\text{Vol}(M(\mathcal{M}, \alpha_1, \dots, \alpha_p)) = \int_{\mathcal{H}_{\text{reg}}} \text{Vol}(M(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta .$$

Proof. First, by [BL99, Thm. 5.20], $M_{g,n}(\alpha_1, \dots, \alpha_p)$ contains at least one element for which the stabiliser under the diagonal action of $\text{SU}(n)$ is $Z(\text{SU}(n))$ if $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}^0$ and either $p \geq 3$, $p = 1$ and $g(\mathcal{M}) = 1$, or $g(\mathcal{M}) \geq 2$. Remark that the volume form dt on $\mathcal{A}$ introduced in the previous subsection is such that

$$\text{Vol} \left(\left\{ (t_1, \dots, t_n) \in \mathbb{R}^n \mid \sum_{i=1}^n t_i = 0, \max_{1 \leq i, j \leq n} t_i - t_j < 1 \right\} \right) = 1 .$$

By (6.1), we have that $\phi(-t) = \widetilde{\phi(t)}$ and $\text{Jac}(\phi(t)) = 1$ for all $t \in \mathcal{A}$. Hence, doing the change of variable $\theta = \phi(t)$ yields

$$\text{Vol}(M(\mathcal{M}, \alpha_1, \dots, \alpha_p)) = \frac{1}{k} \int_{\mathcal{H}_{\text{reg}}^0} \text{Vol}(M(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta ,$$

where $k = 1$ if $\widehat{\mathcal{M}}$ is connected and $k = n$ otherwise. It remains to replace the integration on $\mathcal{H}_{\text{reg}}^0$ by the integration on $\mathcal{H}_{\text{reg}}$ in the case where $\widehat{\mathcal{M}}$ is connected. For all $t \in [0, 1/n)$, $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{\text{reg}}^0$ and $\theta \in \mathcal{H}_{\text{reg}}^0$, by (6.2)

$$\text{Vol} [M_{\text{U}(n)}(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, t \cdot \theta, \widetilde{t \cdot \theta})] = \text{Vol} [M_{\text{SU}(n)}(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})] .$$

Therefore, if $\widehat{\mathcal{M}}$ is connected,

$$\begin{aligned} & \frac{1}{k} \int_{\mathcal{H}_{\text{reg}}^0} \text{Vol}(M(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta \\ &= n \int_0^{1/n} \left(\int_{\mathcal{H}_{\text{reg}}^0} \text{Vol}(M_{\text{U}(n)}(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, t \cdot \theta, \widetilde{t \cdot \theta})) \Delta(\theta)^2 d\theta \right) dt \\ &= \int_{\mathcal{H}_{\text{reg}}} \text{Vol}(M_{\text{U}(n)}(\widehat{\mathcal{M}}, \alpha_1, \dots, \alpha_p, u, \tilde{u})) \Delta(u)^2 du , \end{aligned}$$

where we use that the change of variable $(u_1, \dots, u_n) = (\theta_1 + t, \dots, \theta_{n-1} + t, -\sum_{i=1}^{n-1} \theta_i + t) =: \phi(\theta_1, \dots, \theta_{n-1}, t)$ yields $\text{Jac}(\phi) = n$. $\square$

6.3. Proof of Theorem 2.6. The proof of Theorem 2.6 is then a deduction from the previous results and the previous construction on differential structures.

Proof of Theorem 2.6. The proof is done by induction on $N = 3g + p$, where $N \geq 3$. If $N = 3$, the condition $p + 2g - 2 \geq 1$ implies that $p = 3$ and $g = 0$. The surface is therefore the three-holed sphere and the result is given by Theorem 4.16. Suppose $N > 3$ and let S be a surface of genus g with p points removed. Let $\mathcal{T}$ be a surface constructed in Section 2.3. Then $\mathcal{T}$ is obtained either by gluing two edges of a connected surface $\mathcal{T}'$ or by gluing one edge of a connected surface $\mathcal{T}'$ to the edge of an equilateral triangle T .

In the first case, $\mathcal{T}'$ is a flat surface associated to a surface $\mathcal{M}'$ with genus $g - 1$ and $p + 2$ points removed. Let $\alpha_1, \dots, \alpha_p \in \mathcal{H}_{reg}$ with $\sum_{i=1}^p |\alpha_i| \in \mathbb{N}$. By applying (6.2) and Theorem 6.2, we have

$$\begin{aligned} Z_{g,p}(\alpha_1, \dots, \alpha_p) &= Z_{g,p}(\hat{\alpha}_1, \dots, \hat{\alpha}_p) = \int_{\mathcal{H}_{reg}} \text{Vol}(M(\mathcal{M}', \hat{\alpha}_1, \dots, \hat{\alpha}_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta \\ &= \int_{\mathcal{H}_{reg}} \text{Vol}(M(\mathcal{M}', \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) \Delta(\theta)^2 d\theta. \end{aligned}$$

Let us denote by $c_{g,p} = \frac{c_{0,3}^N}{n^{2g+p-3}}$ where $c_{0,3}$ is given in Theorem 2.6. Since $3(g - 1) + p + 2 < N$, by induction

$$\text{Vol}(M(\mathcal{M}', \alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})) = \frac{c_{g-1,p+2}}{\Delta(\theta)\Delta(\tilde{\theta}) \prod_{j=1}^p \Delta(\alpha_j)} \sum_{G \in \mathcal{G}(g-1,p+2)} \text{Vol}[\text{HONEY}_{\mathcal{T}'}^G(\alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})],$$

and thus by Proposition 5.4,

$$\begin{aligned} Z_{g,p}(\alpha_1, \dots, \alpha_p) &= \frac{c_{g-1,p+2}}{\prod_{j=1}^p \Delta(\alpha_j)} \int_{\mathcal{H}_{reg}} \sum_{G \in \mathcal{G}(g-1,p+2)} \text{Vol}[\text{HONEY}_{\mathcal{T}'}^G(\alpha_1, \dots, \alpha_p, \theta, \tilde{\theta})] \frac{\Delta(\theta)^2}{\Delta(\theta)\Delta(\tilde{\theta})} d\theta \\ &= \frac{c_{g-1,p+2}}{\prod_{j=1}^p \Delta(\alpha_j)} \sum_{G \in \mathcal{G}(g,p)} \text{Vol}[\text{HONEY}_{\mathcal{T}}^G(\alpha_1, \dots, \alpha_p)], \end{aligned}$$

where we used the fact that $\Delta(\theta) = \Delta(\tilde{\theta})$ on the second equality. Since $c_{g-1,p+2} = c_{g,p}$, one gets the result. In the second case, $\mathcal{T}$ is obtained by gluing a surface $\mathcal{M}'$ with genus g and $p - 1$ points removed and a triangle T associated to a three-holed sphere. Let $\widehat{M} = \mathcal{M}' \cup T$ be the corresponding disconnected surface. Then, by (6.2) and Proposition 6.2,

$$\begin{aligned} Z_{g,p}(\alpha_1, \dots, \alpha_p) &= Z_{g,p}(\hat{\alpha}_1, \dots, \hat{\alpha}_p) \\ &= \frac{1}{n} \int_{\mathcal{H}_{reg}^0} \text{Vol}(M(\mathcal{M}', \hat{\alpha}_1, \dots, \hat{\alpha}_{p-2}, \theta)) \text{Vol}(M(T, \tilde{\theta}, \hat{\alpha}_{p-1}, \hat{\alpha}_p)) d\theta. \end{aligned}$$

Set $s = -\sum_{i=1}^{p-2} |\alpha_i|$. By (6.2),

$$\text{Vol}(M(\mathcal{M}', \hat{\alpha}_1, \dots, \hat{\alpha}_{p-2}, \theta)) = \text{Vol}(M(\mathcal{M}', \alpha_1, \dots, \alpha_{p-2}, \theta + s))$$

and

$$\text{Vol}(M(T, \tilde{\theta}, \hat{\alpha}_{p-1}, \hat{\alpha}_p)) = \text{Vol}(M(T, \tilde{\theta} - s, \alpha_{p-1}, \alpha_p)).$$

Since the map $\theta \mapsto \theta - s$ is volume preserving,

$$\begin{aligned} \int_{\mathcal{H}_{reg}^0} \text{Vol}(M(\mathcal{M}', \hat{\alpha}_1, \dots, \hat{\alpha}_{p-2}, \theta)) \text{Vol}(M(T, \tilde{\theta}, \hat{\alpha}_{p-1}, \hat{\alpha}_p)) d\theta \\ = \int_{\mathcal{H}_{reg}^s} \text{Vol}(M(\mathcal{M}', \alpha_1, \dots, \alpha_{p-2}, \theta)) \text{Vol}(M(T, \tilde{\theta}, \alpha_{p-1}, \alpha_p)) d\theta. \end{aligned}$$

By induction,

$$\text{Vol}(M(\mathcal{M}', \alpha_1, \dots, \alpha_{p-2}, \theta)) = \frac{c_{g,p-1}}{\Delta(\theta) \prod_{i=1}^{p-2} \Delta(\alpha_i)} \sum_{G \in \mathcal{G}(g,p-1)} \text{Vol}[\text{HONEY}^G(\alpha_1, \dots, \alpha_{p-2}, \theta)],$$

and

$$\text{Vol}(M(T, \tilde{\theta}, \alpha_{p-1}, \alpha_p)) = \frac{c_{0,3}}{\Delta(\tilde{\theta})\Delta(\alpha_{p-1})\Delta(\alpha_p)} \sum_{G \in \mathcal{G}(0,3)} \text{Vol}[\text{HONEY}^G(\tilde{\theta}, \alpha_{p-1}, \alpha_p)],$$

and thus, by Proposition 5.3,

$$\begin{aligned} Z_{g,p}(\alpha_1, \dots, \alpha_p) &= \frac{c_{g,p-1} c_{0,3}}{\prod_{i=1}^p \Delta(\alpha_i)} \frac{1}{n} \int_{\mathcal{H}_{reg}^s} \sum_{G_1 \in \mathcal{G}(g,p-1), G_2 \in \mathcal{G}(0,3)} \text{Vol}[\text{HONEY}^{G_1}(\alpha_1, \dots, \alpha_{p-2}, \theta)] \\ &\quad \text{Vol}[\text{HONEY}^{G_2}(\tilde{\theta}, \alpha_{p-1}, \alpha_p)] \frac{\Delta(\theta)^2}{\Delta(\theta)\Delta(\tilde{\theta})} d\theta \\ &= \frac{c_{g,p}}{\prod_{i=1}^p \Delta(\alpha_i)} \sum_{G \in \mathcal{G}(g,p)} \text{Vol}[\text{HONEY}^G(\alpha_1, \dots, \alpha_p)] \end{aligned}$$

since $c_{g,p} = \frac{c_{g,p-1} c_{0,3}}{n}$. □

7. YANG-MILLS PARTITION FUNCTION ON COMPACT ORIENTED SURFACES

The goal of this section is to prove Corollary 2.8, which provides an explicit volume formula for the marginal Yang–Mills partition function on an oriented surface of genus g with prescribed non-degenerate holonomies (up to conjugation) along a finite collection of disjoint loops. As shown in [Lév03], this partition function depends only on the prescribed conjugacy classes and on the areas of the connected components delimited by the loops.

To each disjoint loops configuration $\mathcal{L}$, recall the construction of the oriented graph $\mathcal{T}(\mathcal{L}) = (V, E)$ from Section 2.5 :

- the set V of vertices of $\mathcal{T}(\mathcal{L})$ is the set of connected components of $S \setminus \bigcup_{i=1}^p \Gamma_i$. Each vertex $v \in V$ is labelled (A_v, g_v) where A_v is the area of the corresponding connected component and g_v is its genus.
- For $v_1, v_2 \in V$, there is a directed edge e from v_1 to v_2 if v_1 and v_2 are boundary components of a loop, and we write $v = s(e)$. We label the corresponding oriented edge e by α_e , with the condition that $\alpha_{\bar{e}} = 1 - \alpha_e$ if $\bar{e}$ is the oriented edge (v_2, v_1) .

In this section, for $T \geq 0$ and $x, y \in \mathcal{H}$, let us denote by $k_T(x, y)$ the kernel of the unitary Dyson Brownian motion on $\mathcal{H}$ at time T . This kernel has an explicit expression for $x, y \in \mathcal{H}_{reg}$ as

$$(7.1) \quad k_T(x, y) = \frac{\Delta(y)}{\Delta(x)} \det [p_T(2\pi x_i, 2\pi y_j), 1 \leq i, j \leq n] ,$$

where p_T is the heat kernel of the Brownian motion on the circle $\mathbb{R}/2\pi\mathbb{Z}$.

Lemma 7.1 (Partition function of a cylinder). *Let $T > 0$, $\alpha_1 \in \mathcal{H}$, $\alpha_2 \in \mathcal{H}_{reg}$. Then, the Yang–Mills partition function associated to a cylinder of volume T with holonomies α_1, α_2 on its boundaries is*

$$Z_{0,2,T}(\alpha_1, \alpha_2) = \frac{1}{\Delta(\alpha_2)^2} k_T(\alpha_1, \alpha_2) .$$

Proof. This is a consequence of [Lév03, Proposition 4.2.4 and (5.3)]. $\square$

Proposition 7.2 (Volume formula for Yang–Mills partition function). *Let $g \geq 0, p \geq 1$ be integers, $(\alpha_1, \dots, \alpha_p) \in \mathcal{H}^p$ and $T > 0$. Assume that loops $\gamma = \{\gamma_1, \dots, \gamma_p\}$ associated to $\alpha_1, \dots, \alpha_p$ enclose contractible domains $D_1, \dots, D_p$ and that $S \setminus \bigcup_{i=1}^p D_i$ has volume T . Then, the Yang–Mills partition function on $S \setminus \bigcup_{i=1}^p D_i$ with holonomy α_i around γ_i is given by*

$$Z_{g,p,T}(\alpha_1, \dots, \alpha_p) = \int_{\mathcal{H}} Z_{g,p}(u_1, \dots, u_p) \prod_{i=1}^p k_{\frac{T}{p}}(1 - u, \alpha_i) \prod_{\ell=1}^p du_i .$$

Proof. Recall that the partition function is invariant by area preserving diffeomorphisms. Let us introduce a curve γ enclosing the curve γ_p such that the area of the domain enclosed by γ_p and γ is T' , with $0 < T' < T$. Then, by the Markov property of the Yang–Mills partition function, see [Lév03, Proposition 5.1.2] or [Wit92, Eq. (2.63)],

$$Z_{g,p,T}(\alpha_1, \dots, \alpha_p) = \int_{\mathcal{H}} Z_{g,p,T-T'}(\alpha_1, \dots, \alpha_{p-1}, u) Z_{0,2,T'}(1 - u, \alpha_p) \Delta(u)^2 du .$$

Choosing $T' = T(1 - \frac{1}{p})$ and doing the same for the $p - 1$ other curves yields

$$Z_{g,p,T}(\alpha_1, \dots, \alpha_p) = \int_{\mathcal{H}} Z_{g,p}(u_1, \dots, u_p) \prod_{i=1}^p k_{\frac{T}{p}}(1 - u_i, \alpha_i) \prod_{i=1}^p du_i ,$$

where $Z_{g,p}(u_1, \dots, u_p) = Z_{g,p,0}(u_1, \dots, u_p)$ is the the volume of flat connections with holonomies $u_1, \dots, u_p$ on the boundary, see [Sen03]. $\square$

Proof of Corollary 2.8. The proof is done by induction on the number p of loops on S . Let α_p be the holonomy around the oriented loop γ_p of S , and denote by $\tilde{S}$ the surface obtained by cutting S along γ_p .

If $\tilde{S} = S_1 \sqcup S_2$ is disconnected, with γ_1 being positively oriented on S_1 , then by [Lév03, Proposition 5.1.3].

$$\text{YM}_{\mathcal{L}}(\alpha_1, \dots, \alpha_p) = \text{YM}_{\mathcal{L}_1}(\alpha_{i_1}, \dots, i_s, \alpha_p) \text{YM}_{\mathcal{L}_2}(\alpha_{j_1}, \dots, \alpha_{j_s}, 1 - \alpha_p) ,$$

where $\mathcal{L}_1$ is the loop configuration given by S_1 together with the loops $\gamma_{i_1}, \dots, \gamma_{i_r}, \gamma_p$ which lie on S_1 , and $\mathcal{L}_2$ is the loop configuration given by S_2 and the remaining loops $\gamma_{i_1}, \dots, \gamma_{i_r}, \gamma_p$ lying on S_2 .

If $\tilde{S}$ is connected and $\tilde{\mathcal{L}}$ is the loop configuration on $\tilde{S}$ obtained by keeping the loops $\gamma_1, \dots, \gamma_{p-1}$ and the two copies γ_p^+, γ_p^- obtained by cutting along γ_p , then by [Lév03, Proposition 5.4.3],

$$\mathrm{YM}_{\mathcal{L}}(\alpha_1, \dots, \alpha_p) = \mathrm{YM}_{\tilde{\mathcal{L}}}(\alpha_1, \dots, \alpha_p, 1 - \alpha_p) .$$

Iterating on all the loops yields

$$\mathrm{YM}_{\mathcal{L}}(\alpha_1, \dots, \alpha_p) = \prod_{v \in V} Z_{g_v, d_v, T}(\alpha_{e_1^v}, \dots, \alpha_{e_{d_v}^v}) ,$$

where for each $v \in V$, $e_1^v, \dots, e_{d_v}^v$ are the oriented edges starting from v in the graph $\mathcal{T}(\mathcal{L})$ constructed from $\mathcal{L}$ above. Using Proposition 7.2 yields the result. $\square$

REFERENCES

- [AB83] M. F. Atiyah and R. Bott. “The Yang-Mills equations over Riemann surfaces”. In: *Philos. Trans. R. Soc. Lond., Ser. A* 308 (1983), pp. 523–615. DOI: [10.1098/rsta.1983.0017](#).
- [Bel08] P. Belkale. “Quantum generalization of the Horn conjecture”. In: *J. Am. Math. Soc.* 21.2 (2008), pp. 365–408. DOI: [10.1090/S0894-0347-07-00584-X](#).
- [BL99] J.-M. Bismut and F. Labourie. “Symplectic geometry and the Verlinde formulas”. In: *Surveys in differential geometry. Vol. V: Differential geometry inspired by string theory*. Boston, MA: International Press, 1999, pp. 97–311.
- [Buc+16] A. S. Buch, A. Kresch, K. Purbhoo, and H. Tamvakis. “The puzzle conjecture for the cohomology of two-step flag manifolds”. In: *J. Algebr. Comb.* 44.4 (2016), pp. 973–1007. DOI: [10.1007/s10801-016-0697-3](#).
- [Don83] Simon K Donaldson. “A new proof of a theorem of Narasimhan and Seshadri”. In: *Journal of Differential Geometry* 18.2 (1983), pp. 269–277.
- [FT24] Q. François and P. Tarrago. *Positive formula for the product of conjugacy classes on the unitary group*. 2024. arXiv: [2405.06723 \[math.RT\]](#).
- [Jef94] L. C. Jeffrey. “Extended moduli spaces of flat connections on Riemann surfaces”. In: *Math. Ann.* 298 (1994), pp. 667–692. DOI: [10.1007/BF01459756](#).
- [JK98] L. C. Jeffrey and F. C. Kirwan. “Intersection theory on moduli spaces of holomorphic bundles of arbitrary rank on a Riemann surface”. In: *Ann. Math. (2)* 148.1 (1998), pp. 109–196. DOI: [10.2307/120993](#).
- [KL22] A. Kassel and T. Lévy. “Determinantal probability measures on Grassmannians”. In: *Ann. Inst. Henri Poincaré D, Comb. Phys. Interact.* 9.4 (2022), pp. 659–732. DOI: [10.4171/AIHPD/152](#).
- [Kly98] A. A. Klyachko. “Stable bundles, representation theory and Hermitian operators”. In: *Sel. Math., New Ser.* 4.3 (1998), pp. 419–445. DOI: [10.1007/s000290050037](#).
- [KT99] A. Knutson and T. Tao. “The honeycomb model of $GL_n(\mathbb{C})$ tensor products. I: Proof of the saturation conjecture”. In: *J. Am. Math. Soc.* 12.4 (1999), pp. 1055–1090. DOI: [10.1090/S0894-0347-99-00299-4](#).
- [KTW04] A. Knutson, T. Tao, and C. Woodward. “The honeycomb model of $GL_n(\mathbb{C})$ -tensor products II: Puzzles determine facets of the Littlewood-Richardson cone”. In: *J. Am. Math. Soc.* 17.1 (2004), pp. 19–48. DOI: [10.1090/S0894-0347-03-00441-7](#).
- [Lév03] T. Lévy. *Yang-Mills measure on compact surfaces*. Vol. 166. 790. 2003. DOI: [10.1090/memo/0790](#).
- [MW99a] E. Meinrenken and C. Woodward. “Moduli spaces of flat connections on 2-manifolds, cobordism, and Witten’s volume formulas.” In: *Advances in geometry*. Boston, MA: Birkhäuser, 1999, pp. 271–295.
- [MW99b] Eckhard Meinrenken and Chris Woodward. “Moduli spaces of flat connections on 2-manifolds, cobordism, and Witten’s volume formulas”. In: *Advances in geometry*. Springer, 1999, pp. 271–295.
- [NS65] M. S. Narasimhan and C. S. Seshadri. “Stable and unitary vector bundles on a compact Riemann surface”. In: *Ann. Math. (2)* 82 (1965), pp. 540–567. DOI: [10.2307/1970710](#).
- [Sen03] A. N. Sengupta. “The volume measure for flat connections as limit of the Yang-Mills measure”. In: *J. Geom. Phys.* 47.4 (2003), pp. 398–426. DOI: [10.1016/S0393-0440\(02\)00229-2](#).
- [Ver88] E. Verlinde. “Fusion rules and modular transformations in 2D conformal field theory”. In: *Nucl. Phys., B* 300 (1988), pp. 360–376. DOI: [10.1016/0550-3213\(88\)90603-7](#).
- [Wit92] E. Witten. “Two dimensional gauge theories revisited”. In: *J. Geom. Phys.* 9.4 (1992), pp. 303–368. DOI: [10.1016/0393-0440\(92\)90034-X](#).